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# Belated stock returns for green innovation under carbon emissions trading market☆

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ABSTRACT

Carbon trading is an important market mechanism to achieve carbon neutrality. This study explores the possible impact of carbon markets on the stock market performance of listed companies using data from 2013 to 2022 in China's carbon trading pilot regions. Using the event shock of delayed trading in the Chinese carbon market, we attempt to answer the question of the role of green innovation hidden under the compliance event. Results show that a 1% increase in carbon market turnover leads to an average decrease of CNY 0.123 in the company's stock price. Large-scale companies that have been listed for a short time and have poor green innovation capabilities are more vulnerable to the carbon market. Under delayed trading, firms with high green innovation capability will be profitable. By contrast, profitability is not reflected in low green innovation firms. Companies with low green innovation can reduce stock market performance owing to the carbon market's undersupply situation. Our study reveals the stock performance of different carbon trading entities under delayed trading, providing a realistic basis for firms to choose green innovation while helping to improve carbon trading market dynamics.

## 1. Introduction

Climate change is one of the most serious environmental challenges facing the world today, and the annual increase in carbon emissions is an important factor contributing to global climate change (Friedlingstein et al., 2011; Rode et al., 2021). The carbon trading market emerged under the framework of the Kyoto Protocol and became an important initiative for energy saving and emission reduction (Nordhaus, 2006; Bauer and Rudebusch, 2021; Peñasco et al., 2021). Existing studies mainly focus on the trading mechanism, economic effects, and market performance of the carbon emission trading market (Fowlie et al., 2012; Kollenberg and Taschini, 2016; Trinks et al., 2020; Bolton and Kacperczyk, 2021). Since 2013, China's seven pilot carbon markets have been launched one after another. As of 31 December 2021, the cumulative turnover of carbon emission allowances in the seven pilot carbon markets reached 483 million tons, with a turnover of CNY 8.622 billion.[1] The largest potential carbon market is China, and studying the impact of its

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[1] The data come from the energy economy forecast and outlook research report. This is a series of research reports completed by the Energy and Environmental Policy Research Center and the research team of Beijing University of Technology.

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carbon trading on the stock market performance of carbon-intensive companies can help us understand the behavioral decisions of emission control companies and reveal the potential transmission mechanism of the carbon market, potentially affecting financial performance and ultimately improving the efficiency of the carbon market and guiding most of the trading behavior of companies.

The link between decarburization and financial markets has been a hot topic in environmental economics, and many cutting-edge developments have emerged, such as the study of the synergy between carbon markets and economic development and the relevance of carbon trading to financial risk (Bolton and Kacperczyk, 2021; Volz et al., 2021). Academic studies on the correlation between carbon markets and company stock performance are scarce, and the findings are inconsistent (Hartzmark and Sussman, 2019; Krueger et al., 2020; Azar et al., 2021; Bauer and Rudebusch, 2021). The carbon emission trading system is an important environmental regulation policy under the market economy system (Abrego and Perroni, 1999), but it may not necessarily be beneficial for enterprises. (Choi et al., 2020; Krueger et al., 2020; Van den Bremer and Van der Ploeg, 2021). On the one hand, high carbon-emitting industries are heavily influenced by carbon trading prices, and stock returns will decline owing to increased carbon market trading costs (Bushnell et al., 2013; Nguyen and Phan, 2020). Conversely, carbon trading increases companies' costs but will ultimately be borne by consumers (Fabra and Reguant, 2014). On the other hand, in terms of net profits from carbon allowances, most companies participating in carbon trading have shown good stock market performance (Carlson et al., 2000; Bolton and Kacperczyk, 2021; Hickey et al., 2021). Developments in stock and credit markets can also affect technological innovation (Hsu et al., 2014). Exploring the impact of the carbon trading market on stock market performance is therefore necessary.

The Porter hypothesis suggests that appropriate environmental regulations can promote green innovation (Porter and Linde, 1995). However, the impact of environmental regulations on green innovation is the result of a series of trade-offs for companies (Bhattacharya et al., 2017). On the one hand, environmental regulations may lead to "crowding-out effects" on green innovation. Under the assumptions of neoclassical economics, environmental regulations can be seen as an additional constraint. Companies aiming for profit maximization may be unable to achieve their original profits under these additional environmental constraints and may even exit the market if no viable solutions exist within environmental constraints. Environmental regulations also have an impact on corporate financial decision-making (Bose et al., 2021; Dang et al., 2022). Market-oriented environmental regulations provide companies with more choices, but companies still need to incur certain costs to meet the requirements, squeezing out funds for green innovation (Murphy et al., 1993; Shleifer and Vishny, 1994; Feichtinger et al., 2005; Qiu et al., 2018). On the other hand, environmental regulations may also have "compensatory effects" on green innovation. Carbon emission trading markets allow innovative companies to operate without emission constraints, with surplus credits that can be sold, leading to compensation. Therefore, solely considering crowding-out effects may underestimate the policy effects of carbon emission trading.

We choose the Chinese market as the subject of this study, which is an ideal setting for this study. First, the environmental regulation involved in the China carbon exchange is a highly politicized constraint, which ensures compliance activities of emission control companies. Large transactions will be made at the end of the year for companies with high carbon emissions to ensure compliance. This delay in trading at the end of each year has led to a specific pattern in the Chinese carbon market and provided a shock of events, helping us explore green innovation's role in carbon trading activities. Second, the Chinese carbon market has adopted a gradual liberalization approach, with eight pilot regions gradually liberalized between 2013 and 2017, and daily carbon trading data for each region are recorded in detail on local exchanges. The pilot regions have undergone thorough preparations before the market's opening to ensure that carbon emissions trading in the pilot regions is carried out.

To gain a deeper understanding of the status of China's carbon trading market, we collect data on carbon emissions, carbon market transactions, and total corporate quotas in seven pilot areas[2] in China from 2014 to 2019 for analysis. If enterprises fail to meet their emission reduction targets, they can fulfill their emission reduction responsibilities by purchasing insufficient quotas on the market. As shown in Fig. A1, the orange and gray colors represent the total annual trading turnover of the carbon market and the quota volume of the carbon market in the current year. At present, China's carbon market mainly involves key emission units in the power generation industry. Carbon emissions from 2014 to 2019 are relatively large, the share of carbon market transactions and carbon quotas is relatively small, and company participation is insufficient. The herd effect (Zhou et al., 2022), or fulfillment transactions mode, reduces the initiative of enterprises in the carbon market and causes losses in carbon market efficiency. This study therefore explores the impact of carbon market trading on the stock market performance of carbon-intensive companies and the role of corporate green innovation in this context based on the pattern of delayed trading in China.

We devote considerable attention to explaining that the annual compliance supervision of the pilot cities has led to the impact of the compliance period in the Chinese carbon market. China's Ministry of Ecology and Environment claimed in 2022 that the emission control firms do not actively perform the contract, they are only active in the convergence period of the performance deadline, and the market trading turnover is close to the quota gap of enterprises. In the subsequent analysis, we use the event study approach to discuss the impact of fulfillment-based trading in detail and supplement the robustness of the research design. More deeply, we consider the potential seasonal factors that may exist in the event study method. For one thing, we use the heating differences between the north and the south to test. For another, machine learning methods are also used to explore whether the monthly characteristics are the most important feature vectors that affect the increase of carbon trading turnover. After ensuring the robustness of the method, we use it to explore the effects of green innovation. We utilize the impact of the surge in carbon transactions at the end of the year caused by the performance-based trading model to deeply analyze the impact of green innovation on the trading behavior of controlled enterprises. Does the regular increase in carbon transactions at the end of the year necessarily affect companies' economic or environmental

[2] Considering that Fujian began the carbon trading pilot in 2017, Fig. A1 does not include Fujian Province.

performance? This discussion is the entry point for our study. We are also trying to find commonalities in different industries. Whether companies at the industry's leading level of green innovation have better stock price performance than weak companies under the background of increased trading turnover in the carbon market. The influence of the carbon market on the stock market to some extent mirrors investor expectations. Our findings offer a fresh research perspective for deciding between decarbonization innovation and participation in the carbon market, going beyond the consideration of their direct costs.

First, we develop a well-established empirical strategy that demonstrates the negative impact of carbon trading on stock price markets. We find that the shorter the listing time, the stronger the profitability; the weaker the capital-intensiveness, the greater the sensitivity. The level of green innovation is divided according to the top 25% of the industry. If the company is at a fraction of the industry's green innovation level, the negative impact of controlling emissions will be minimal or even non-existent. Second, we conduct an in-depth analysis using policy shocks during the compliance period. The mechanism analysis of green innovation implies that, for regions with significant compliance period shocks, the stock price of firms with high green innovation capacity increases at the end of the year, with insignificant effects in firms with weak green innovation capacity. In the cost–benefit analysis, firms with a high level of green innovation will increase the level of firm revenues owing to the surge in carbon market transactions, which will nearly enhance the firm's performance in the stock market. Spot carbon market volatility may also lead to short-term volatility in share prices by affecting investor expectations. We use >10,000 observations of carbon emission corporate data and get corroboration. The stronger the level of green innovation, the lower the number of corporate carbon emissions, which nearly increases operating income.

We have contributed to the relevant literature in several important ways. First, compared with the existing literature, we are the first to explore the impact of carbon trading on stock prices in carbon-intensive industries using daily data from all carbon exchanges traded in China. On the basis of the Porter hypothesis, we explore the potential causal relationship of these problems and uncover the black box of the company's green development decisions. Compared with studies of corporate activity in the stock market (Fang et al., 2014; Tan et al., 2020), we extend the channel of the impact of carbon markets on financial markets and enrich the related literature. For a placebo test, we replace the original sample with a non-carbon-intensive industry. We also redefine the scope of industries based on power generation and downstream industries, trying to explore carbon market-sensitive industries.

Second, this study is the first to exploit the special characteristics of delayed trading by using compliance period shocks in a pilot region of China to control for changes in carbon trading turnovers owing to unobservable factors. We explore the economic impact of transactions on different green companies from the market phenomenon of delayed transactions. This approach avoids endogenous variation in high-frequency daily data and facilitates further causality determination and mechanism exploration. We extend the literature on the role of local policy orientation on innovation (Tian and Xu, 2022). Our findings also help establish a link between the carbon and stock markets, providing evidence for green innovation incentives for emission-controlling firms.

Third, from the cost–benefit analysis perspective, we attempt to answer whether emission-controlling companies choose to innovate on their own or buy carbon allowance from the carbon market. We provide new evidence of the positive impact of green innovation on emission control companies. Companies with strong green innovation will see their stock prices rise when demand in the carbon market is high, whereas companies with weak green innovation will see poor stock market performance. The study results can provide quantitative support for the carbon market, thus helping the policy designation of carbon control companies and guiding their green innovation.

Our research also provides relevant references for countries implementing market-based environmental policies. First, the carbon trading market can motivate enterprises to innovate green innovation and generate benefits through stock price. Many studies discuss the effects of different means of environmental policy (Baldursson and Von der Fehr, 2004; Hao et al., 2022), and we provide strong support for market-based environmental policies. Second, although the carbon trading market can reward green innovative enterprises, many companies trade carbon allowances according to production conditions at the end of the year, which may lead to fluctuations in corporate financial markets (Martin et al., 2014; Chen et al., 2023). We find that the fluctuation range of the stock price is about [0.072, 1.319], accounting for [0.384%, 7.033%] of the average stock price. Therefore, when implementing carbon trading policies, being alert to financial risks is necessary. The parameters estimated in this paper also provide a reference for preventing financial risks.

The rest of the paper is organized as follows. Section 2 reviews some pertinent existing research. Section 3 details the data and presents our identification strategy. Section 4 details our main results. Section 5 performs a mechanistic analysis. Section 6 concludes.

## 2. Literature review

The emission trading scheme is seen as a cost-effective way to reduce CO2 emissions in addition to a carbon tax. Coase (2013) first proposed that, if property rights are clearly defined, people can effectively choose the most advantageous way to transact and minimize transaction costs, thus solving various problems through transactions. If carbon emissions are treated as an attributed right, the cost of reducing emissions to society can be minimized by trading this right in a free market. Calel (2020) found that the European carbon market encourages regulated companies to carry out more low-carbon patent applications and R&D expenditures, but it will not promote the reduction of output carbon intensity in the short term. Acemoglu (2023) constructed an endogenous technology growth model and found that if innovation is completely guided by the market, distortions in innovation will occur and social welfare will undergo enormous losses, which are due to the externality of pollutant emissions.

China's emissions growth had been largely led by strong economic growth. With the establishment of pilot cities in China's carbon market in recent years, research on carbon market trading in China has received increasing attention. China is the world's second-largest emitter of greenhouse gases and the world's largest supplier of certified emission reductions (CERs) to the primary market. Therefore, studying the impact of China's carbon emissions market performance on the economy is particularly important. Aghion

et al. (2016) found that when faced with higher tax-included fuel prices, firms tend to innovate more in clean technologies. Chen et al. (2021) extended the study of carbon emissions to the impact on corporate green innovation and found that the establishment of carbon markets promoted corporate green innovation. Under the background of China's carbon emissions trading pilot, Almond and Zhang (2021) found that the establishment of a carbon emissions market has contributed to improvements in air quality. Moshirian et al. (2021) found that stock market liberalization improves the efficiency of capital allocation in the innovation process of enterprises, thereby promoting productivity growth.

The relationship between carbon emissions trading and financial markets has been more popular in recent years (Oestreich and Tsiakas, 2015; Cai and Lontzek, 2019; Peñasco et al., 2021). Oberndorfer (2009) analyzed the impact of the European Union Emissions Trading Scheme (EU ETS) on the stock market by applying econometric analysis for the first time, showing that the EU ETS price has a significant impact on the stock performance of electricity companies. Gerlagh and Liski (2018) found that the returns to capital and climate investments were no longer equal, leading to a significant increase in emissions taxes compared to a benchmark with equal returns. Fabra and Reguant (2014) measured the pass-through of emission costs, implying that emission costs are fully passed on to electricity prices, and firms have little incentive to adjust price increases after cost shocks.

Many studies have examined the impact of environmental regulation on firm performance (Jorgenson and Wilcoxen, 1990; Larch and Wanner, 2017; Brucal et al., 2019; Singh and Chanda, 2021). From a theoretical perspective, the carbon market influences other capital markets through the return spillover channel and the volatility spillover channel (Kodres and Pritsker, 2002; Acharya and Pedersen, 2005). Karpf et al. (2018) analyzed the European carbon trading system and found that industry participants turned to local connections and financial intermediaries to participate in the market due to the lack of well-identified trading institutions. Siemroth and Hornuf (2023) indicated that investors prefer to have a positive impact on the environment and meet that impact by investing in green projects. Chu et al. (2021) explored how firms alter their green innovation activities and strategies in response to environmental spills near their headquarters.

In summary, the current studies analyzing the carbon market and the stock market mostly treat policy implementation as a quasi-natural experiment and lack studies on the stock market performance of the power generation industry and related emission control industries. Although studying the situation of carbon markets in the EU and other regions has implications for China's carbon market and the stock market performance, they are not contextualized in the context of China's pilot policies. The literature also mostly explores the relationship between the stock and carbon markets based on the VAR, GARCH, and autoregressive models, but less in-depth analysis is conducted. Under the endogeneity problem of potential two-way causality of two variables, how to deeply explore the influence mechanism becomes the focus of this paper.

We first refined the study subject and selected carbon-sensitive industries in China's carbon pilot regions, indicating that a large number of pollutant emissions accompany the industry's main business. Second, based on China's highly politicized context of environmental regulation, we utilized the compliance period event that the company had to face near the end of the year as an exogenous shock. We also combined green innovation data of listed companies and tried to explore the impact of green innovation on the listing performance of emission control companies through heterogeneity analysis and the event study approach.

## 3. Data and empirical strategy

### 3.1. Transaction data of carbon market

We combine the policy context of China's carbon emissions trading pilot regions, which are summarized and displayed in Table 1. On October 29, 2011, the General Office of the National Development and Reform Commission issued the Notice on the Pilot Project of Carbon Emission Trading, agreeing that seven provinces and cities, namely, Beijing, Shanghai, Tianjin, Chongqing, Hubei, Guangdong, and Shenzhen, should carry out pilot projects of carbon emission trading. At the end of 2013, pilot trading of carbon emission rights was carried out in various regions, with the pilot trading in Hubei launched in April 2014, Chongqing in June 2014, and Fujian in December 2016. The industries in these pilot areas involved carbon trading cover power generation, iron and steel, chemical, thermal,

**Table 1**
Descriptive statistics for China pilot exchange data.

| Policy area | Pilot time | Performance times | Sample size | Average price of carbon emissions | Sample Date |
|---|---|---|---|---|---|
| Shenzhen | 2013.6.18 | 8 | 76,807 | 36.976 | 2013.8.5–2022.6.30 |
| Shanghai | 2013.11.26 | 8 | 66,421 | 36.170 | 2013.12.19–2022.6.30 |
| Beijing | 2013.11.28 | 8 | 75,298 | 61.938 | 2013.11.28–2022.6.30 |
| Guangdong | 2013.12.19 | 8 | 131,338 | 29.556 | 2013.12.19–2022.6.30 |
| Tianjin | 2013.12.26 | 8 | 5749 | 24.399 | 2013.12.26–2022.6.30 |
| Hubei | 2014.4.2 | 7 | 44,914 | 26.071 | 2014.4.2–2022.6.30 |
| Chongqing | 2014.6.19 | 7 | 11,525 | 18.842 | 2014.6.19–2022.6.23 |
| Fujian | 2016.12.22 | 5 | 14,712 | 22.030 | 2017.1.9–2021.1.13 |

Notes: This table presents the chronological order of policy pilots. As of 2022, China has eight pilot regions (including municipalities directly under the central government) for carbon market trading. Shenzhen is a special economic zone, and although it is part of Guangdong Province, the carbon emissions trading pilot policies are separate and independent. Performance times refers to the total number of carbon trading compliance cycles experienced up to the year 2022. The carbon market price unit is CNY per ton. The number of sample observations depends on the differences in the number of listed companies in the pilot cities and the differences in the time windows of the pilot regions.

manufacturing, and airline transportation. Each pilot region's daily carbon market trading data are obtained from the China Stock Market & Accounting Research Database (CSMAR). The carbon market trading data include the daily opening price, closing price, highest price, lowest price, average price, volume, and turnover of carbon trading.

The development and reform commissions and local governments in the pilot regions report good compliance with the carbon market in each region, with compliance rates reaching 100% in each area since 2015 (no published data are available for Hubei and Chongqing pilot cities). Table 2 presents the pilot regions' basic data according to the carbon market's operational events. It contains the market opening time, the amount of compliance experienced, the sample size, the observations' distribution, and the transactions' average price. Shenzhen is the earliest region to carry out carbon market trading. Shenzhen is part of Guangdong Province, and the number of observations reflects the length of time the pilot region has been open and the number of corresponding listed companies. Guangdong also has the highest number of observations. The price of carbon emission rights varies significantly in different regions. In the Beijing carbon market, the trading price is CNY 61.938 per ton, the highest price exchange. The data we collected on carbon trading coincided with the time of listing in the pilot cities, reflecting the data's reliability.

### 3.2. Stock market trading data and control variables

The financial data came from the CSMAR, which provides information on all stock transactions from the Shanghai and Shenzhen Stock Exchanges since their inception. We first match the stock codes to individual company information, including the stock market trading data, market type, trading status, industry type, founding date, initial listing date, provincial location, city, and main business. We also match the data to filter out companies whose locations were not in the pilot area. Subsequently, we pair relevant variables in firm-level financial statement information, including operating cycle, total assets, total liabilities, Tobin's q, total operating revenue, operating cost, operating profit, capital intensity, current insensitive ratio, accounts receivable to revenue, cash ratio, and the lagged binary variable reflecting whether a stock closed higher than its opening price on the preceding trading day. Considering the three- and five-factor models (Fama and French, 1997; Fama and French, 2015), we also add market risk premium factors, market value factors, the difference in the yield of high book-to-market ratio combination, and low book-to-market ratio combination, which commonly influence stock prices. The market risk premium factor is the difference between the daily market return rate of cash dividend reinvestment and the daily risk-free interest rate (the central bank announced the benchmark interest rate for fixed deposits in March).

Variable-oriented regression results may be produced, so we replace the core variables in the benchmark regression by replacing the stock's daily closing price with the daily highest stock price in the robustness test. Stock transactions are high-frequency daily data, and the time series type data may be affected by the stock price of the previous trading day. Thus, we generated the previous period data of the stock and added it to the model as a control variable. In the heterogeneity analysis, the variation of carbon emissions among companies with different levels of green innovation was considered. Thus, we use the China Research Data Service Platform, which is a high-quality, open, and platform-based comprehensive data platform for economic, financial, and business research in China, to obtain

**Table 2**
Summary statistics of key variables.

| Variables | Obs. | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|
| Clsprc | 341,215 | 18.756 | 33.192 | 1.173 | 1441.020 |
| CarbonTA | 341,215 | 11.090 | 3.385 | 0 | 18.421 |
| LnAssets | 341,215 | 22.454 | 1.594 | 16.727 | 28.651 |
| lnDebt | 341,215 | 21.301 | 2.000 | 12.032 | 28.292 |
| Tobin's_Q | 341,215 | 2.109 | 2.077 | 0.694 | 78.620 |
| lnProfit | 341,215 | 18.787 | 1.998 | 9.326 | 25.609 |
| lnIncome | 341,215 | 20.705 | 2.177 | 4.483 | 28.693 |
| LR | 341,215 | 2.197 | 2.536 | 0.100 | 64.071 |
| CR | 341,215 | 0.630 | 1.476 | −0.065 | 52.734 |
| RAL | 341,215 | 0.443 | 0.196 | 0.013 | 1.140 |
| RARI | 341,215 | 0.800 | 3.468 | 0.000 | 195.346 |
| LnCI | 341,215 | 1.759 | 1.554 | −1.408 | 13.871 |
| Fluctuate (Lag) | 341,215 | 0.597 | 0.491 | 0 | 1 |
| Gpo | 273,756 | 54.581 | 208.236 | 0 | 2263 |
| Gpa | 273,756 | 77.746 | 305.417 | 0 | 3499 |
| Listing_time | 341,215 | 10.726 | 8.125 | 0 | 31 |
| RiskPremium | 341,215 | 0.000 | 0.016 | −0.100 | 1.075 |
| SMB | 341,215 | 0.000 | 0.008 | −0.091 | 0.059 |
| HML | 341,215 | 0.000 | 0.006 | −0.044 | 0.056 |

Notes: This table shows descriptive statistics of the main variables. Clsprc (unit: yuan) represents the daily closing price of the company's stock, CarbonTA represents the turnover of carbon traded in the region on the trading day (logarithmically processed), Assets represents the company's assets, Debt represents the company's debt level, Profit means corporate profit, LR is the liquid ratio, CR is the cash ratio, RAL is the balance sheet ratio. RAL is the ratio of assets to liabilities, RARI is the ratio of accounts receivable to income, CI is capital intensity, and Fluctuate is a lagged binary variable reflecting whether a stock closed higher than its opening price on the preceding trading day. Gpo is the number of green patents obtained, Gpa is the number of green patent applications. Listing_time shows the length of listing time of the listed company, and the unit is year. RiskPremium means Market risk premium factor, SMB is the difference in yield between small-cap stock portfolio and large-cap portfolio and HML is the difference between the yield of the high book-to-market ratio combination and the low book-to-market ratio combination.

the annual green patent applications and green patent acquisitions of listed companies as the analysis of heterogeneity and mechanism.

The listed companies in this study include industries closely related to carbon emissions, such as electricity, heat production and supply, metal mining, oil and gas extraction, metal processing, oil processing, railroad transportation, electrical machinery, and chemical manufacturing. Our selection of industries was based on the main range of industries covered by the pilot region. China's carbon market has a national carbon offset mechanism for Chinese Certified Emission Reduction (CCER), which implies that enterprises controlling carbon emissions quantify and certify the emission reduction effect and register the emission reductions in the national greenhouse gas voluntary emission reduction trading registry system. Specifically, the CCER market allows non-focused emission control firms to enter and provides a trading platform to sell their audited voluntary emission reductions, creating a correlation between firms. Some pilot regions involve industries that include only high-carbon emitting firms in key power generation industries; thus, we include only firms related to power generation industries in the robustness tests.

### 3.3. Descriptive statistics of the sample

Table 2 shows descriptive statistics for the main variables. The core explanatory variable is the stock's daily closing price. It has a mean price of CNY 18.756, a minimum price of CNY 1.173, and a maximum price of CNY 1441.02. This implies a significant difference in stock prices. We use daily high-frequency carbon and stock market data. High-frequency trading data are expected to lead to timely volatility. Thus, we do not consider ex-rights, ex-dividend, or dividend scenarios, meaning we have not adjusted the share price. Considering the large turnover and uneven distribution of carbon trading credits, we logarithmized the turnover of carbon trading credits. Tobin's q-value has a mean of 2.109 over the sample period, indicating that most listed firms are overvalued. We also combine green patent data from listed firms based on the China Research Data Service Platform. Companies filed 77 green patents per year and received an average of 54 patents per year during the sample period. Using the listing time of listed companies, we calculate the length of time companies were listed up to 2022. During the sample period, the company was listed as late as 2022, while its earliest listing occurred >31 years ago. The mean of the market risk premium factor, the difference between the returns of the small-cap and large-cap portfolios, and the difference between the returns of the high and low book-to-market ratios are all approximately equal to zero, with small standard deviations.

### 3.4. Empirical strategy

This section details the identification strategy used to estimate the causal impact of carbon market trading turnover on stock prices in the stock market. The identification is based on the carbon market context, where large emission types of listed companies, such as power plants, must trade in the carbon market to purchase when their emissions are exceeded. To realize this aim, we estimated the following model (Oestreich and Tsiakas, 2015; Monasterolo and de Angelis, 2020):

$$Clsprc_{it} = \beta CarbonTA_{ct} + Controls_{it} + \rho_c + \theta_t + \lambda_m + \epsilon_{ict} \quad (1)$$

where $\text{Clsprc}_{it}$ is the closing price of stock trading for company $i$ in date $t$. $\text{CarbonTA}_{ct}$ refers to the pilot regional level mean in the carbon trading turnover of pilot region $c$ at day $t$, which we logarithmized. $Controls_{it}$ is a vector of individual information about the company. We obtain a set of control variables that may affect stock prices using data from the financial statements of listed companies. It contains company assets, liabilities, current ratio, corporate Tobin's q value, cash ratio of operating profit, gearing ratio, accounts receivable to revenue ratio, capital intensity, and the lagged binary variable reflecting whether a stock closed higher than its opening price on the preceding trading day. We also add a series of factors that may affect stock prices, including market risk premium factors, the difference in yield between small-cap stock portfolio and large-cap stock portfolio, and the difference in yield between high book-to-market ratio and low book-to-market ratio, the difference between the yield of the high-profit stock portfolio and the low-profit portfolio, the yield difference between the low-investment ratio stock portfolio, and the high-investment ratio stock portfolio. These indicators are weighted according to the circulating market value, and we also conduct a robustness test according to the total market value weight, which corresponds to Table A2. To mitigate potential confounding effects at the enterprise level, we meticulously select control variables across multiple dimensions. After accounting for corporate heterogeneity, this study further incorporates firm-level fixed effects and conducts a microscopic analysis of carbon emissions at the enterprise level. We could not include all control variables, just to ensure no omission of variables relevant to the core explanatory variable, while ensuring robust identification. The results are shown in Columns 6 and 7 of Table A3, and all the results are significant and consistent with the baseline.

This research has some special considerations. Our transaction data are concentrated at the end of the year. Therefore, a correlation may be found between specific months, so the standard errors are clustered at the industry-by-month level. We set year-by-month-fixed effects ($\theta_t$) and province-fixed effects ($\rho_c$), implying that identification comes from within-year-by-month and within-province variations. Considering the heterogeneous trend of stock prices across industries, we included industry-fixed effects ($\lambda_m$) in the benchmark regressions. Our parameter of interest is $\beta$, which means the change in stock trading price owing to the change in trading turnover in the carbon market.

For the model setting, we explore the following possible problems. First, although we have included a series of fixed effects and control variables to ensure exogenous changes in carbon trading credits, some concerns that changes in carbon trading credits may be affected by unobserved effects. Hintermann (2010) argued that market forces on behalf of electricity and heat companies may influence EU ETS allowance prices. Based on the event study approach, we introduced event shocks in the compliance period to explore whether the increase in carbon trading turnovers at the end of the year leads to a change in stock prices. Kothari and Warner (2007)

reinforced the importance of using stratified samples to examine event study statistical properties. Therefore, in Section 5.1, we explore the robustness of the setting and method of the event study approach and elaborate on the potential mechanism.

Second, the day-to-day variation of stock trading prices is affected by many confounding factors, and we try to capture the daily influence of stocks by adding date-fixed effects. For example, since COVID-19, stock prices have been subject to shocks from the rebound of the epidemic. We include year-by-month fixed effects in the baseline regressions to control for variation in a specific year across months within the sample period. We also exclude stock price fluctuations owing to industry variability. Stock prices are potentially influenced by the previous period, which leads to a bias in the estimated coefficients. We include the stock price of the previous trading day as a control variable, and the results are not significantly affected. To address nationwide or regional-level shocks, we consider province-by-year fixed effects. This accounts for regional policies and shocks enacted by provinces in different years. We further consider city-by-year fixed effects, which capture adjustments made by cities in different years, such as the initiation of infrastructure projects and changes in environmental requirements.

Third, we are concerned about pseudo-regressions in the model setup, so we performed a sample swap. Combined with the carbon market trading context, we select the relevant industries in the pilot cities not restricted from carbon emissions as the study subjects and run the regression based on the existing model. The results indicate that the change in the carbon market trading turnover does not change the stock price of unrelated industries, implying that worrying about the results because of pseudo-regressions is unnecessary.

Finally, the identification depends on the changes in the stock trading of carbon emission-intensive companies driven by carbon market trading. Due to data limitations, we cannot obtain the details of the carbon transactions of listed companies in the carbon trading market, so we use the high-frequency daily carbon trading data in the pilot area as a variable to measure the supply and demand relationship in the carbon market. In the model regression, the company's fixed effect has not been added due to our fear that the company's heterogeneous effects may be diluted, such as the carbon emission quota or emission control technology of different emission control companies. Facing the problem of potential identification deviations, we make supplementary treatment. The carbon emission data disclosed by listed companies from 2018 to 2022 well reflect the company's emission control behavior and indirectly reflect the company's supply and demand in the carbon market. Considering the company's fixed effect, we conduct an expanded analysis.

## 4. Results

### 4.1. Baseline results

To explore the reflection of the trading turnover of the carbon emission trading market on the stock market, we conduct a benchmark regression with a series of robustness tests. Table 3 shows the results of the benchmark regression. In Column 1, we perform univariate regressions and find a significant negative relationship between carbon market trading turnover and stock prices of carbon emission-sensitive firms without adding any controls. We add year-by-month-fixed effects and province-fixed effects in Column 2 and additional industry-fixed effects in Column 3. In Column 4, we add data on the characteristics of listed companies, which is also the preferred specification of our study.

Table 3 shows that all results are significantly negative and statistically significant, indicating that changes in the daily trading turnover of the carbon market will affect stock prices. In Column 4, the coefficient is −0.123, which suggests that, for every 1% increase in the daily trading turnover of the carbon market, the stock price of the carbon-sensitive industry in that region decreases by CNY 0.123. In other words, an increase in the trading turnover of carbon emissions rights by one unit will cause the stock price to fall by 0.072 (=0.123/(e−1)). Considering that the average stock price is CNY 18.756 during the sample period, the stock price decline accounts for about 0.384% (=0.072/18.756) of the average stock price. We further consider a daily volume change of one standard deviation (3.385). At this time, this will cause the stock price to drop by CNY 0.242 (=0.123/(e−1)*3.385), which accounts for about 1.290% (=0.242/18.756) of the average stock price. Given that carbon trading is often concentrated at the end of the year, this will lead to a sudden increase in trading turnover. We assume a more serious situation: the trading turnover of the day rose from the lowest to the highest. Currently, the daily volume change is 18.421. This will result in a loss of CNY 1.319 (=0.123/(e−1)*18.421) in the stock

**Table 3**
Baseline results.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Clsprc | Clsprc | Clsprc | Clsprc |
| CarbonTA | −0.055*** | −0.170*** | −0.169*** | −0.123*** |
| | (0.014) | (0.053) | (0.049) | (0.042) |
| Year×month FEs | N | Y | Y | Y |
| Province FEs | N | Y | Y | Y |
| Industry FEs | N | N | Y | Y |
| Controls | N | N | N | Y |
| Observations | 422,916 | 422,916 | 422,916 | 341,215 |
| R-squared | 0.000 | 0.028 | 0.066 | 0.123 |

Notes: No fixed effects with control variables were added in Column 1. Year-by-month-fixed effects and provincial fixed effects were added in Column 2. We added industry-fixed effects in Column 3 and control variables in Column 4. Standard errors were clustered at the industry-by-month level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

price, accounting for 7.033% (=1.319/18.755) of the average stock price. This represents a huge shadow for the stock market. We give the range of stock price fluctuations caused by different changes. We assume that the maximum scale of carbon trading will not change. Then, the range of stock volatility brought about by carbon trading is [0.072, 1.319], accounting for [0.384%, 7.033%] of the average stock price. Due to the surge in China's carbon trading scale at the end of the year, a year-end shock will have a huge impact on stocks price.

### 4.2. Robustness checks

This subsection presents a series of robustness analyses by combining the nature of the carbon market with the stock market. Table 4 shows the results of replacing the settings of the preferred specification. The price of a stock may be influenced by the price of the previous business day, creating an expectation among investors. Column 2 includes the stock price in the previous period as a control, which is not significantly different from the results of the benchmark regression. In Column 3, the stock's daily closing price is replaced with the daily maximum price for replacement. Moreover, the daily trading turnover of the carbon market is replaced with the daily trading price and daily trading turnover, and the regression results are shown in Columns 4 and 5. Column 6 includes control for market capitalization. We log the market capitalization and found that the effect of market capitalization on stock price is significantly positive with a substantial coefficient value, while the effect of carbon market trading turnover on stock price remains significantly negative. Column 7 adds the variable of listing time. All results of Table 4 imply that our results are robust.

The identification depends on the stock market's performance in the face of changes in supply and demand in the carbon market. In the benchmark regression, we do not add the fixed effect at the enterprise level. For one thing, since the pilot of China's carbon market pilot in 2013, we cannot obtain carbon transaction data at the enterprise level. Instead, we use daily carbon market data as useful information. Enterprise-level decision-making or emission control resources may also transmit potential impact effects. For example, enterprises with low carbon emissions and high green innovation will master the initiative of carbon trading. If the enterprise fixed effect is added directly to the model, it will dilute the regression results we are interested in. We propose three robust expansions for this problem. First, we combine the carbon emission data at the company level with the stock market data and add the enterprise fixed effect to confirm our conjecture. The results are shown in Table A1. Regardless of whether the enterprise fixed effect is added, high-carbon emission enterprises will face a greater impact on carbon trading. Second, after distinguishing the heterogeneity of innovation ability, we add enterprise fixed effect in the event study. Third, based on the characteristics of China's fulfillment-based trading, we expand the event research on the impact of the surge in trading turnover at the close of the year, and the method design is discussed in detail.

Table 5 shows the results of a battery of robustness tests. In Column 2, the tail for the core explanatory variable and daily carbon market transactions was shrunk, and the results remain consistent with the benchmark regression results. In Columns 3 and 4, the scope of industries is narrower, considering only metal manufacturing and processing, electricity, heat production and supply, coal and oil mining, electrical machinery and equipment manufacturing, and other related industries as carbon-emission sensitive industries and chemical manufacturing, air transportation, and rail transportation as non-core sensitive industries. The delineation lines are based on the fact that the set industries in our pilot regions vary in scope, mainly around the power generation industry, with the

**Table 4**
Robust check: change the preferred specification.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| | Clsprc | Clsprc | Hiprc | Clsprc | Clsprc | Clsprc | Clsprc |
| CarbonTA | −0.123*** | −0.145*** | −0.128*** | | | −0.087** | −0.188*** |
| | (0.042) | (0.033) | (0.044) | | | (0.039) | (0.042) |
| lnCarbonTV | | | | −0.085** | | | |
| | | | | (0.040) | | | |
| lnAvgprice | | | | | −2.713*** | | |
| | | | | | (0.361) | | |
| lnMV | | | | | | 14.697*** | |
| | | | | | | (1.611) | |
| List year | | | | | | | −0.875*** |
| | | | | | | | (0.078) |
| Year×month FE | Y | Y | Y | Y | Y | Y | Y |
| Province FE | Y | Y | Y | Y | Y | Y | Y |
| Industry FE | Y | Y | Y | Y | Y | Y | Y |
| Clsprc(lag) | N | Y | N | N | N | N | N |
| Controls | Y | Y | Y | Y | Y | Y | Y |
| Observations | 341,215 | 340,799 | 341,215 | 344,393 | 344,196 | 341,215 | 341,215 |
| R-squared | 0.123 | 0.434 | 0.123 | 0.123 | 0.124 | 0.143 | 0.152 |

Notes: This table shows the results of the regressions set under changing the preferred specification. Column 2 adds the stock price of the previous period as a control variable. In Column 3, the core explanatory variable uses the highest daily stock price. In Column 4, the total turnover of daily carbon transactions is replaced by the number of carbon transactions, which is logarithmic. In Column 5, the core explanatory variable is replaced with the average price of carbon market transactions. In Columns 6 and 7, the stock market value and listing time are added to the preferred model. FEs means fixed effects, and the following table is the same. All models' standard errors are clustered at the industry-by-month level and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

**Table 5**
A battery of robust test.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| | Preferred spec. | Winsorized | Core industry | Non-core industry | Drop ST* | Add FEs | Price volatility |
| CarbonTA | −0.123*** | −0.121*** | −0.263*** | 0.078* | −0.120*** | −0.130*** | −0.013*** |
| | (0.042) | (0.042) | (0.049) | (0.045) | (0.043) | (0.046) | (0.003) |
| Year×Month FEs | Y | Y | Y | Y | Y | Y | Y |
| Province FEs | Y | Y | Y | Y | Y | Y | Y |
| Industry FEs | Y | Y | Y | Y | Y | Y | Y |
| Type FEs | N | N | N | N | N | Y | N |
| Controls | Y | Y | Y | Y | Y | Y | Y |
| Observations | 341,215 | 341,215 | 207,387 | 133,828 | 335,486 | 341,215 | 274,356 |
| R-squared | 0.123 | 0.123 | 0.129 | 0.247 | 0.123 | 0.124 | 0.106 |

Notes: This table presents a series of robust tests. In Column 2, CarbonTA is scaled up or down by 2.5%. We divided the industry scope in more detail, Column 3 is the core industry sample, and Column 4 is the regression result of the non-core industry sample. In Column 5, ST stocks are excluded. FEs means fixed effects, and the following table is the same. In Column 6, we added the fixed effect of carbon market product types. In column 7, the interpreted variable is the short-term stock volatility, which is measured by the difference between the highest and lowest prices of the stock of the day. Standard errors are clustered at the industry-by-month level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

downstream industry component not participating in the emissions trading market. Thus, we narrow the scope of our study by subdividing the industries. In addition, we convert the three factors in the control variable into five factors and replace the calculation method weighted by the circulating market value with the total market value-weighted. The robustness test is shown in Table A2.

The results show that the higher the degree of participation in the carbon market, the higher the carbon market transactions and the lower the corporate stock prices, affecting twice as much as the full sample. Companies in non-carbon market core industries are not significantly affected. These results imply that the more correlated with carbon trading, the more vulnerable the firms are to carbon market trading, whereas the downstream industries were not significantly affected. In Column 5, we exclude ST stocks to prevent the effect of abnormal stock price fluctuations on the results, and the final result remains significantly negative and not significantly different from the benchmark regression. We also attempt to control for shocks at different levels by replacing fixed effects in Column 6 of Table 5.

In Column 7, we replace the interpreted variable with the daily fluctuation of the stock, which is measured by the difference between the highest and lowest prices of the stock of the day. Table 5 implies the increase in the trading turnover of the carbon market will lead to a decrease in short-term stock volatility. From the perspective of the size of the coefficient, the degree of short-term fluctuation of stocks caused by changes in carbon trading turnover is small compared with stock prices. A 1% increase in carbon market turnover leads to an average decrease of CNY 0.013 in the company's stock volatility. For short-term traders and speculators, minimal price fluctuations may diminish profit potential, potentially hampering investors investing. In Table A3, observations from the stock market crash period were excluded. We also replace a variety of fixed effects and change the cluster stand error with robustness tests.

Table 6 shows the results of the placebo test. If the placebo test does not pass, it indicates that stock prices are not affected through carbon market trading but through unobserved effects. We select four industries unrelated to carbon trading: livestock, accommodation, business services, and retail. The firm's location is all pilot cities in the carbon market, and the model is set consistent with the benchmark regression. Column 1 is the sample total for the four industries. Columns 2 to 5 are the sub-sample regressions for the above

**Table 6**
Placebo test.

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| | Clsprc | Clsprc | Clsprc | Clsprc | Clsprc |
| CarbonTA | −0.032 | 0.020 | −0.053 | 0.068 | −0.028 |
| | (0.030) | (0.013) | (0.030) | (0.079) | (0.031) |
| Year×Month FEs | Y | Y | Y | Y | Y |
| Province FEs | Y | Y | Y | Y | Y |
| Industry FEs | Y | Y | Y | Y | Y |
| Controls | Y | Y | Y | Y | Y |
| Observations | 76,389 | 1918 | 1835 | 34,867 | 37,767 |
| R-squared | 0.346 | 0.972 | 0.949 | 0.385 | 0.426 |

Notes: This table is a placebo test. We used industries unrelated to carbon trading as the subject of the study, mainly including four categories: livestock, accommodation, business services, and retail trade. We replaced the sample according to the setting of the baseline. It is a good placebo test because the above companies are in carbon trading pilot areas. Column 1 is the total sample of unrelated listed companies, Column 2 is the sample of listed companies in the livestock sector, Column 3 is the accommodation sector, Column 4 is the business services sector, and Column 5 is the retail sector. The models are set up consistent with the baseline regressions, with FEs referring to fixed effects, as is the case in the table below. Standard errors are grouped at the industry-by-month level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

four industries. All our results show that changes in carbon market trading turnover in pilot cities do not affect stock prices in industries unrelated to carbon emissions.

### 4.3. Heterogeneity analysis

We find that a rise in carbon market turnover will lead to a decline in the stock prices of carbon-intensive enterprises. More deeply, to identify sensitive groups, we conduct the heterogeneous analysis according to the characteristics of the enterprise. First, from the perspective of the company's listing time, older enterprises may be better able to deal with the negative impact of carbon trading. Adelino et al. (2017) connected new and old enterprises with investment creation models. Coad et al. (2016) explored the relationship between innovation and enterprise development of companies of different ages. Second, from the perspective of profit, companies with weak profitability may be more sensitive to emissions control. Homroy (2023) studied the enterprise greenhouse gas emission on profitability, implying that the emission reduction improved the company's profitability through information advantages. In the research on green credit policies on corporate carbon emissions, Lee et al. (2022) explored the heterogeneity of the impact from a capital-intensive perspective. Finally, the different level of innovation has brought a great differentiated stock price change on the enterprises in the control of excretion (Carrión-Flores and Innes, 2010). Zhu et al. (2019) found that China's carbon emission trading pilot has increased the low-carbon innovation of enterprises by 5% to 10%. Combined with the above literature, in this section, we tried to carry out a heterogeneous analysis from the four dimensions of the firm's operating time, business status, innovation level, and firm type.

We calculate the length of time the company will be listed as of June 30, 2022. Companies that have been listed for more than eight years are divided into a subsample, and the remainder are companies that have been listed for a short period, which are in their initial development period as a subsample. The eight-year setting aligns with the business cycle of listed companies and is consistent with the sample median of listed companies. In addition to dividing sub-samples by median thresholds, we also divide samples into short and long sub-samples according to the quartile (4 years and 18 years). We also construct the dummy variables separately and used interaction items to explore their heterogeneity. Moreover, we perform some heterogeneity analysis, which includes differences in operating income levels, green innovation capabilities, and firm nature.

Our results show that the increase in carbon market trading turnover for companies with a short-listing time leads to a decrease in stock price, with a more significant impact coefficient that exceeds three times the results of the benchmark regression. This result suggests that an increase in carbon market trading turnover has a more significant negative impact on the stock price of startups. In Column 3, whether the listing age is more than eight years is defined as a dummy. We omit the regression coefficient of the variable and retained the interaction term. The result is still consistent with the sample regression. We changed the median division to the quartile division, and the results are displayed in Table A9, which is consistent with Table 7.

We then divide the sample according to the operating revenue level (sample median). The results are displayed in Columns 4 and 5 of Table 7, where we find that for companies with high operating revenue and large scale, the company's stock is more vulnerable to the carbon market. The coefficient of Column 6 interaction also corresponds to the conclusion that the high-profit group is more likely to be adversely affected by the carbon market. The possible reason is that high-profit companies usually bear greater costs and risks in carbon emissions, which may have a negative impact on their profitability and market competitive position.

Table 8 demonstrates the heterogeneity brought by the capital intensity of firms and the level of green innovation. Capital intensive ratio is expressed as total assets divided by operating income. When the denominator is not published or is zero or less than zero, it is

**Table 7**
Heterogeneity analysis: listing time and profit level.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | S_Time | L_Time | Interaction | L_profit | H_ profit | Interaction |
| CarbonTA | −0.472*** | 0.043 | −0.333*** | −0.035 | −0.139*** | −0.024 |
| | (0.086) | (0.028) | (0.080) | (0.040) | (0.049) | (0.062) |
| CarbonTA#L_T | | | 0.264*** | | | |
| | | | (0.083) | | | |
| CarbonTA#H_P | | | | | | −0.198** |
| | | | | | | (0.087) |
| Year×Month FE | Y | Y | | Y | Y | Y |
| Province FE | Y | Y | | Y | Y | Y |
| Industry FE | Y | Y | | Y | Y | Y |
| Controls | Y | Y | | Y | Y | Y |
| Observations | 156,195 | 185,020 | 341,215 | 170,590 | 170,625 | 341,215 |
| R-squared | 0.136 | 0.297 | 0.148 | 0.236 | 0.206 | 0.124 |

Notes: Column 1 of this table is a sample of enterprises that have been listed for less than eight years (sample median), and Column 2 is for enterprises that have been listed for more than eight years. In Column 3, the L_T variable and the interaction between the core explanatory variable and L_T are added. L_T is the dummy variable of whether the company has been listed for more than eight years. Column 4 is for enterprises with low operating income type, and Column 5 is for enterprises with high operating income type, and the threshold value of operating income is set as the sample median (20.64971). The interaction between the core explanatory variable and whether it is a high operating income is added to the last column. FEs means fixed effects and the following table is the same. Standard errors are clustered at the industry-by-month level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

expressed as the vacancy value. This study uses the capital intensity of 9.09% (quartile) as the threshold to divide the sample. This indicator reflects the amount of assets required for the sales of production units. High-capital-intensive companies include the iron and steel industry, general electronic and communication equipment manufacturing industry, transportation equipment manufacturing industry, and petrochemical industry. Columns 1 to 3 show that firms with the lower capital intensity ratio are affected by lower stock prices. We suggest that firms with higher capital intensity are more capable of withstanding carbon trading and will not experience excessive stock movements when carbon market trading increases. By contrast, firms with lower capital intensity ratios are more vulnerable to the impact of changes in supply and demand trading in the carbon market on the stock market. These enterprises may need more capital investment to adapt to the environmental regulations. Table A4 implies that private companies are more vulnerable to the carbon market than state-owned companies. The two conclusions are consistent. Ahiakpor (1986) pointed out that the capital intensity of private enterprises is lower than that of state-owned enterprises (SOEs). The reason may be that SOEs could become profitable by increasing their capital intensity (Berkowitz et al., 2017), so they can resist the negative impact of carbon emission control on enterprises.

Subsequently, we divide the sample according to the number of applications and acquisitions of the company's green patents (stock and flow level). Among them, the stock of green patents is calculated as an accumulated amount up to the current year, based on 2013 as the reference year, without accounting for depreciation, and the flow level is based on year-by-year increments. Considering industry differences in innovation. We also set the top 25% of companies with green patent licensing in this industry as the group that values green innovation and the bottom 75% of companies ranked as the group that does not value innovation. In terms of the division of green innovation, we did not choose the median, which corresponds to six patent authorizations and eight patent applications. The average number of patent acquisitions in the sample is 55. This exceeds 75% of the quant, resulting in the dominance of extreme values after sample division, so we chose a point of compromise. To ensure the robustness of the results, we also used the stock of green patent applications at the industry level as a supplementary analysis. These results are shown in Table A8. Notably, considering the close relationship between the level of green innovation and the firm's age, we also add the company's listing time to eliminate confounding effects in the relevant research on the level of green innovation.

Combined with the voluntary emission reduction policy, we find that companies with more green patent applications are less affected when facing carbon trading. The impact on the carbon trading market is greater for firms that do not pay attention to green dedication and is greater than the results of the benchmark regression. Therefore, green innovation plays a mitigating role in the adverse effects of carbon markets on firm share prices, providing evidence for the later mechanistic analysis.

## 5. Mechanism analysis

This section first uses the event shock of the approaching compliance period to explore whether the approaching compliance period leads to changes in stock prices using the event study method. Carbon trading may promote green and innovative technologies in companies, increasing their profitability (Chen et al., 2021). It then explores the profitability of green innovation using the impact of carbon markets on the economic performance of firms as a transmission mechanism. Specifically, we analyze the cost-benefit effects of carbon market transactions of listed companies. We collect pollutant emission data from key carbon emission companies since 2018. Considering the availability of data, we provide an in-depth discussion of the transmission of the mechanism through a small sample.

**Table 8**
Heterogeneity analysis: capital intensive ratio and green innovation level.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | H_CI | L_CI | Interaction | L_Gpo | H_Gpo | Interaction |
| CarbonTA | 0.012 | −0.183*** | −0.167*** | −0.056* | 0.058 | −0.173*** |
| | (0.043) | (0.060) | (0.052) | (0.029) | (0.039) | (0.051) |
| CarbonTA#H_CI | | | 0.170** | | | |
| | | | (0.071) | | | |
| CarbonTA#H_GI | | | | | | 0.142** |
| | | | | | | (0.070) |
| Year×Month FE | Y | Y | Y | Y | Y | Y |
| Province FE | Y | Y | Y | Y | Y | Y |
| Industry FE | Y | Y | Y | Y | Y | Y |
| Controls | Y | Y | Y | Y | Y | Y |
| Observations | 85,350 | 255,865 | 341,215 | 205,069 | 68,651 | 341,215 |
| R-squared | 0.297 | 0.115 | 0.124 | 0.163 | 0.423 | 0.126 |

Notes: Column 1 is for the sample with capital intensity >4.5 (the sample median). Column 2 is for the sample with small capital intensity. In Column 3, we added additional variables H_CI (high capital intensity dummy variable) and the interaction of the core explanatory variable and dummy. We divided the samples according to whether the green patent obtained stock is in the top 25% of the industry (H_Gpo), and the regression results are displayed in Columns 4 and 5. The dummy variable and its interaction with carbon trading turnover are added to Column 6. FEs means fixed effects and the following table is the same. Standard errors are clustered at the industry-by-month level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

### 5.1. Compliance period shock

Two challenges were encountered during this period: identifying the intermediate role of green innovation in the impact of the carbon market on the stock market and finding an event shock that can exogenously reflect carbon market trading movements. Using event shocks during the compliance period in Chinese pilot cities, we apply an event study approach to explore the role of green innovation in depth.

Fig. 1 shows the daily trading changes in carbon market trading turnover. The daily trading turnover data are obtained by summing up the data of each pilot city on that day. The time span is from August 5, 2013 to June 30, 2022. The graph shows the year-to-year increase in trading turnover and the cyclical nature of the daily trading turnover changes in the pilot carbon markets. This figure reflects the overall trend of carbon trading. As shown in Table 1, the timing of the pilot area entering the market is different, so the regularity of the change in trading turnover cannot be explored. Since 2017, a different degree of increase in trading turnover at the close of each year has been observed. This could be because given the impact of the compliance period, one will choose to liquidate their carbon credits and buy or sell them in the carbon market. Although Fig. 1 is consistent with the study of Wen (2023), who found that China's carbon trading turnover in November and December will improve significantly, our explanation needs to be careful. For one thing, Fig. 1 does not control the possible seasonal characteristics and area differences. For another, for the cyclical performance shock of carbon trading every year, enterprises may have expected behavior to reduce this fluctuation. For these potential problems, we carry out detailed supplementary tests and explanation of the method.

Figs. 2 and 3 show the variation in trading turnover in different pilot regions. We use October as the start time of the shock and applied the event study method, and the specific model settings are shown below.

$$CarbonTA_{ct} = \alpha + \beta_9 Lead9_{ct} + \beta_8 Lead8_{ct} + \cdots + \beta_2 Lead2_{ct} + \beta_1 Lead1_{ct} + + \gamma_1 Lag1_{ct} + + \gamma_2 Lag2_{ct} + Controls_{ct} + \theta_t + \rho_c + \epsilon_{ct} \quad (2)$$

$$Clsprc_{ict} = \alpha + \beta_9 Lead9_{ct} + \beta_8 Lead8_{ct} + \cdots + \beta_2 Lead2_{ct} + \beta_1 Lead1_{ct} + + \gamma_1 Lag1_{ct} + + \gamma_2 Lag2_{ct} + Controls_{it} + \theta_t + \rho_i + \epsilon_{ict} \quad (3)$$

The dependent variables in Eqs. (2) and (3) are consistent with Eq. (1), and a series of lead and lag are considered relative to the event of the increase in the number of performance transactions at the end of the year. The standard errors are clustered at the year level. $CarbonTA_{ct}$ implies the pilot regional level mean in the carbon transaction turnover of pilot region $c$ at day $t$, which we logarithmized. We regard October as the event of the fulfillment transactions and try to explore whether there is a trend of increasing carbon transactions in pilot areas in October and beyond.

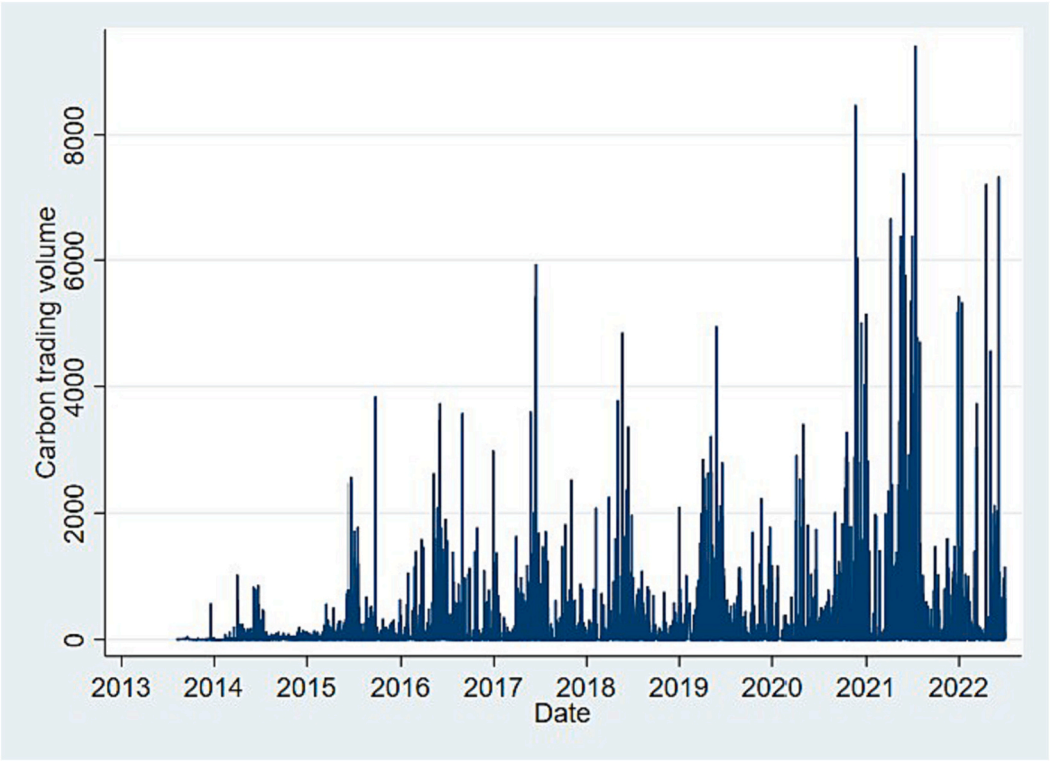



**Fig. 1.** Carbon market trading turnover time-varying chart.
Notes: This figure shows the variation of carbon trading turnover over time for the eight pilot cities. We have summed up the eight pilot regions, with different carbon trading varieties, to obtain the daily total national carbon trading turnover in (100,000 tons). The data covers the period from 2013.8.5 (the first trading day of the first pilot city) to 2022.6.30.

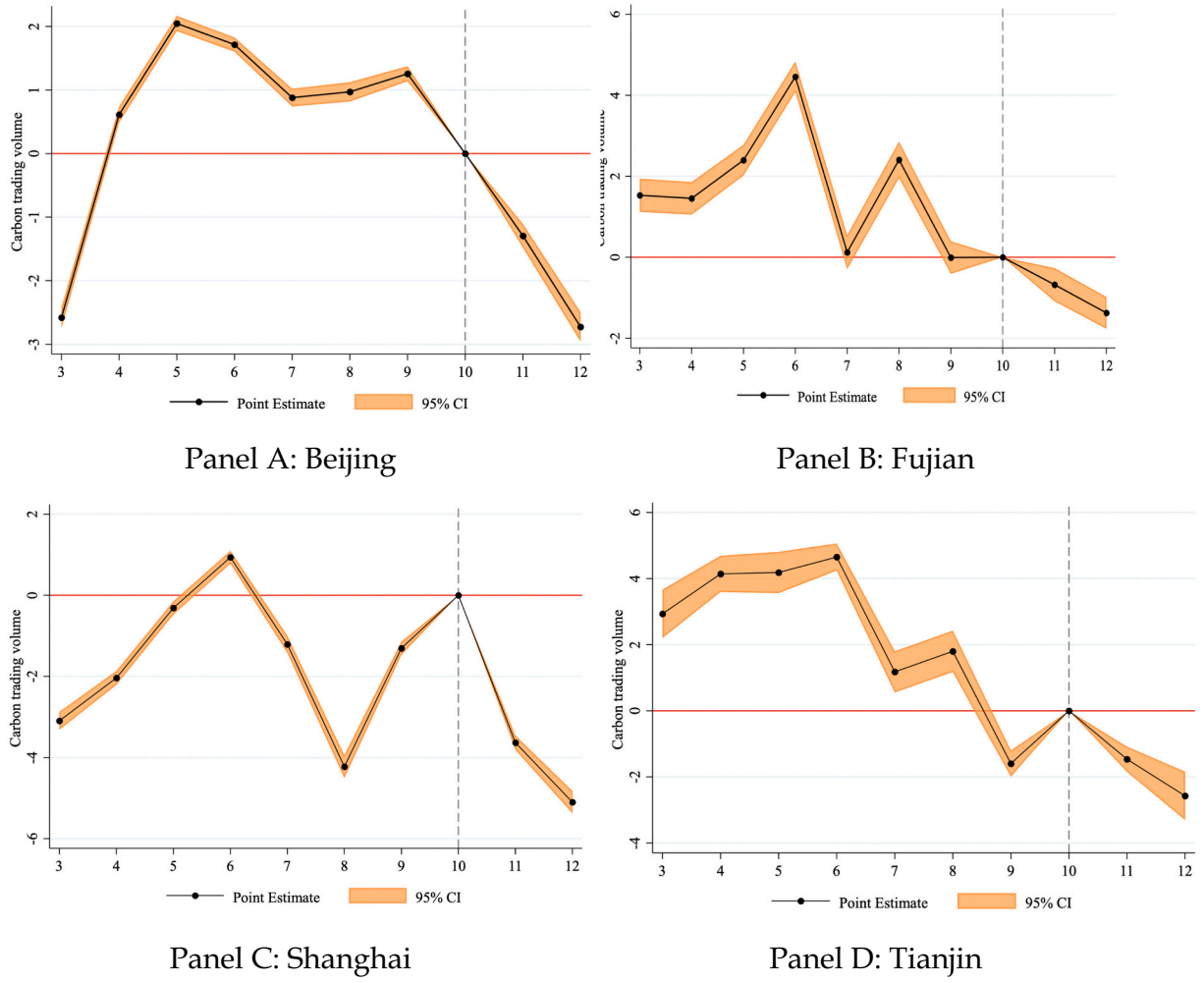



**Fig. 2.** Dynamic trend test for carbon trading turnovers in pilot cities (unaffected).
Notes: This figure shows a dynamic trend test for carbon trading turnovers in the four pilot cities. We used the compliance period as a quasi-natural experiment to explore whether the trading turnover after October was affected by the approaching compliance period, which corresponds to Eq. (2). The x-axis indicates the month. The regression results corresponding to the figure are shown in Table 9.

In Eq. (2), $Controls_{ct}$ includes the industrial producer purchase price index (100 in the same month of the previous year) of the pilot area $c$ at the time of each month. Control variables also include the current value of power generation, industrial added value, thermal power generation, fixed asset investment completion, natural gas production, crude oil production, and gas production. They are closely related to the emission behavior of emission control enterprises (Levinson, 2009; Masnadi et al., 2021). Adding too much control will lead to too few effective samples. We only use the purchase price index of industrial producers as a control variable to test the robustness. Fixed effects for area and year are included as $\theta_t$ and $\rho_c$. To identify the differences in carbon trading turnover in different months, we do not add the fixed effect of the month, so different characteristics may occur between different months due to specific carbon trading patterns. We also explore the event trends of carbon trading in different pilot cities by region. The model is consistent with Eq. (2), but the pilot area fixed effect is removed.

$Clsprc_{ict}$ is the closing price of stock trading for company $i$ in date $t$, which is in pilot city $c$. $Controls_{ct}$ is also consistent with the benchmark regression in Eq. (3), which may affect the company's stock price. We divide the company according to the stock of green patent technology to explore whether the stock performance of enterprises with different levels of green innovation is significantly different under the performance trading model. In Eq. (3), We add firm fixed effect $\rho_i$ and year fixed effect $\theta_t$. In this event study approach, we use October as a marker for the beginning of the compliance period at the end of the year.

We discuss the setting of the event study method in detail. First, utilizing October as a reference involves a fusion of behavioral necessities to ensure year-end compliance, which is a cyclical external shock. Wen (2023) pointed out that in China's carbon trading, transactions mainly converge at the close of the compliance periods, with 13% and 70% of quota transactions in November and December respectively. Furthermore, considering the quarterly nature of corporate financial reporting, the fourth quarter serves as a short-term window for observing corporate performance cycles (Eliasson, 1977; Bathke Jr et al., 1989; Sullivan and von Wachter,

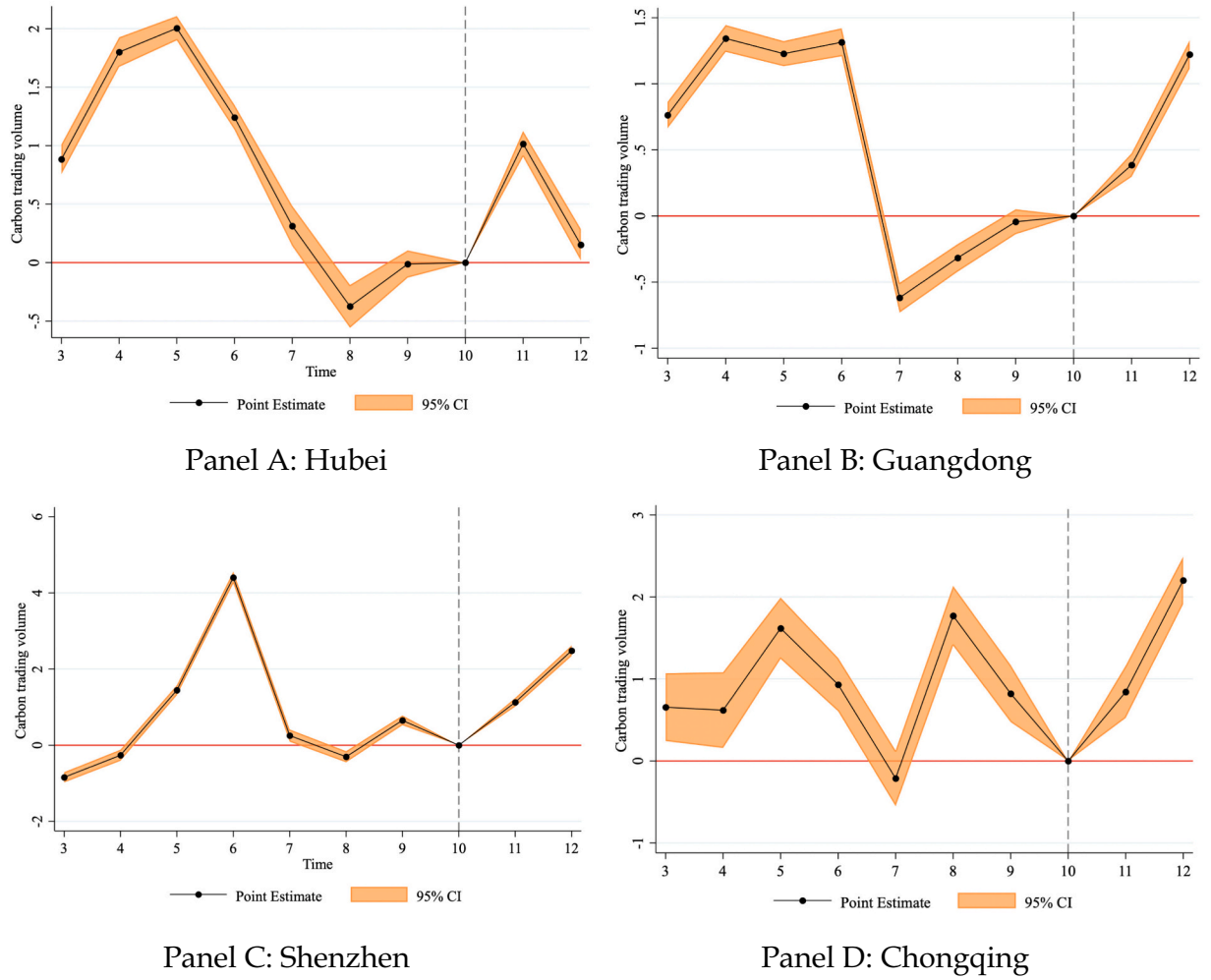



**Fig. 3.** Dynamic trend test of carbon trading turnover in pilot cities (affected).
Notes: This figure shows a dynamic trend test of carbon trading turnovers for the four pilot regions under the compliance period, with the model set up as shown in Eq. (2). We set October and later months as the time when the compliance period occurs. October is used as the reference group. The x-axis indicates the month. These four regions have the same trend at the end of the year, when the carbon trading turnover rises significantly, indicating that the onset of the compliance period has an identical event trend impact for the four pilot regions of Hebei, Guangdong, Shenzhen, and Chongqing. The graph corresponds to Table 9.

2009; Gopinath and Neiman, 2014). Second, the anticipation of the performance impact is considered to ensure the shock is reliable. If the enterprise has expected behavior, the law of performing trading in the carbon market may not exist. Drawing on the research of Kothari and Warner (2007), we stratify the samples in the event study method to explore whether each pilot area has fulfillment-based transactions. Third, in view of the potential seasonal confounding factors for year-end performance events, we learn from the research of Chen et al. (2013) to test the possibility of this seasonal factor from the perspective of heating differences between the North and South. We also use supervised machine learning algorithms to identify the six most important factors affecting carbon trading turnover and found that the transaction is mainly due to regional differences rather than the characteristics of the month.

This study examined the trend of carbon market transactions in different pilot regions during the compliance period by region. We divide the regions with declining trading turnover at the end of the year into a group, indicating that the group whose market trading turnover is not affected by the compliance period. The possible reason is that the carbon trading market entity has made corresponding adjustments according to the expected market trend, and no increase in the transaction turnover will occur due to the performance period at the end of the year. At the same time, we divide the regions with rising trading turnover after October into a group, indicating that pilot regions whose trading turnover will be affected by the compliance period.

The coefficients in Table 9 reflect the changes in the carbon turnover of the pilot area in each month compared with October. To identify the differences between months, we can only set the fixed effect at the annual level, which may not result in an obvious parallel trend in the months before October. If the fixed effect is set at the month or day level, this variance we are interested in is absorbed by the fixed effect. Therefore, we allow for the existence of different monthly changes in carbon trading turnover. Taking Table 9 as an

**Table 9**
Fulfillment transactions event shock.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| | CarbonTA | CarbonTA | CarbonTA | CarbonTA | CarbonTA | CarbonTA | CarbonTA | CarbonTA |
| Lead7 | −3.836*** | 1.539*** | −1.789*** | 4.533*** | 0.896*** | 0.807*** | −1.497*** | −0.165 |
| | (0.071) | (0.175) | (0.104) | (0.487) | (0.082) | (0.060) | (0.084) | (0.206) |
| Lead6 | −0.643*** | 1.461*** | −0.732*** | 5.740*** | 1.813*** | 1.386*** | −0.914*** | −0.202 |
| | (0.062) | (0.172) | (0.081) | (0.378) | (0.085) | (0.054) | (0.083) | (0.218) |
| Lead5 | 0.792*** | 2.405*** | 0.993*** | 5.782*** | 2.017*** | 1.271*** | 0.792*** | 0.797*** |
| | (0.058) | (0.169) | (0.082) | (0.431) | (0.067) | (0.047) | (0.068) | (0.180) |
| Lead4 | 0.459*** | 4.463*** | 2.243*** | 6.250*** | 1.254*** | 1.357*** | 3.751*** | 0.109 |
| | (0.055) | (0.163) | (0.078) | (0.280) | (0.054) | (0.044) | (0.067) | (0.185) |
| Lead3 | −0.375*** | 0.130 | 0.096 | 2.776*** | 0.325*** | −0.575*** | −0.400*** | −1.033*** |
| | (0.060) | (0.156) | (0.097) | (0.312) | (0.063) | (0.046) | (0.067) | (0.192) |
| Lead2 | −0.284*** | 2.414*** | −2.918*** | 3.397*** | −0.362*** | −0.273*** | −0.955*** | 0.950*** |
| | (0.066) | (0.173) | (0.130) | (0.330) | (0.068) | (0.044) | (0.061) | (0.188) |
| Lead1 | −1.255*** | 0.008 | 1.305*** | 1.599*** | 0.013 | 0.044 | −0.653*** | −0.822*** |
| | (0.061) | (0.205) | (0.084) | (0.202) | (0.059) | (0.048) | (0.065) | (0.177) |
| Lag1 | −2.550*** | −0.670*** | −2.327*** | 0.134 | 1.027*** | 0.429*** | 0.475*** | 0.021 |
| | (0.082) | (0.203) | (0.088) | (0.201) | (0.067) | (0.049) | (0.067) | (0.174) |
| Lag2 | −3.983*** | −1.366*** | −3.796*** | −0.975** | 0.165*** | 1.265*** | 1.829*** | 1.381*** |
| | (0.103) | (0.168) | (0.125) | (0.379) | (0.063) | (0.054) | (0.074) | (0.175) |
| Year FEs | Y | Y | Y | Y | Y | Y | Y | Y |
| Controls | Y | Y | Y | Y | Y | Y | Y | Y |
| Observations | 42,412 | 12,885 | 42,948 | 2609 | 31,652 | 85,518 | 52,528 | 6550 |
| R-squared | 0.303 | 0.286 | 0.173 | 0.813 | 0.302 | 0.296 | 0.506 | 0.150 |

Notes: This table shows the temporal trend of carbon market transaction turnover. We treated the approach of the corporate carbon compliance period as a quasi-natural experiment. Columns 1 o 8 represent Beijing, Fujian, Shanghai, Tianjin, Hubei, Guangdong, Shenzhen, and Chongqing. We differed from the benchmark regression in that only year-fixed effects and industry-fixed effects were retained.

example, after accounting for the differences in the year and adding a set of control variables, the carbon trading turnover of the Hubei Stock Exchange in November and December increased by 102.7% and 16.5% compared with October, and the results were significant at the 1% level (Column 5). Figs. 2 and 3 show the changes in carbon market trading turnover in the pilot regions, which are the coefficient visualizations of Table 9. Fig. 2 shows the carbon markets unaffected by the compliance period. Fig. 3 shows the pilot cities where the trading turnover increases due to the arrival of the compliance period. This event study method is modeled in Eq. (2) and the regression results correspond to Table 9. Considering that we add many control variables in Table 9, resulting in too few observations, the regression results of the ninth and eighth stages before the event are removed from the regression. In Table A5, we only use the purchase price index of industrial producers as a control variable to test the robustness.

After a combination with Table 9, among the eight pilot areas, no obvious performance transactions are observed in the four carbon trading markets of Beijing, Fujian, Shanghai, and Tianjin. This means that the volume of carbon trading in the market has not increased significantly in the past two months compared with October. By contrast, the four pilot areas of Hubei, Guangdong, Shenzhen, and Chongqing had a significant impact on performance, and the carbon trading turnover increased in November and December. We then combined the pilot area observations with the same characteristics. We run a total regression on the observations of the two groups and add a regional fixed effect. The regression results are shown in Table A6, and the visual results are shown in Fig. A4. The regression results for the full sample are consistent.

Subsequently, we explore whether there were confounding effects caused by seasonal factors under the impact of the performance period. Chen et al. (2013) used China's Huai River policy as a quasi-natural experiment to explore the impact of continuous pollution, which provides winter heating by providing coal for boilers in cities north of the Huai River but denying heat to the south. Therefore, using the characteristic facts of heating, we divide the sample into the south and the north to explore whether a significant difference exists between the carbon transaction turnover and the carbon emissions of enterprises in the south and the north under the event study method. The results are shown in Tables A11 and A12, and the visual results are shown in Figs. A5 and A6. The results show that whether carbon turnover or corporate carbon emissions are used, no significant difference is found between the southern and northern samples in the event study design, alleviating our concerns about seasonal factors to a certain extent.

We also use the XGBoost (eXtreme Gradient Boosting) integration algorithm to determine the main characteristics that affect the change in carbon trading turnover. First, setting the indicator "Higher" indicates whether the daily carbon transaction turnover is higher than the average annual carbon transaction turnover in the pilot area. Second, we construct the eigenvector and use the random forest to fill the missing values (missing values in a total of 10 indicators, and the number of missing does not exceed half of the observed values). The feature vector includes the dummy variables of region and month, as well as the year, quarter, week, and day of the week that may affect carbon trading, and all the control variables in Eq. (2). Considering that we do not fit the prediction, no adjustment of the parameters is done, and the default parameters "n_estimators" are equal to 100. Finally, through supervised machine learning algorithms, we explore the impact of all feature variables on the classification variable Higher and sort the importance of feature vectors. Table A13 lists the first six factors affecting the variable Higher. We found that one-hot code in the region is the main factor that mainly affects the increase or decrease of trading turnover, rather than the monthly characteristics.

For the sample of observations with fulfillment transaction shocks, we examine the movement of stock trading prices in response to fulfillment transaction shocks. The model setup for our event study approach is shown in Eq. (3). To identify the role of green innovation, we split the sample into high and low-green innovation firms using flow data. Fig. 4 shows stock trading price movements under event shocks. The visualization of the four-panel coefficients in Fig. 4 corresponds to the regression coefficients of Columns 4 and 5 in Table 10. We control for the firm and year fixed effects and add control variables consistent with the benchmark regression. Columns 1 and 3 in Table 10 show that in pilot areas where the carbon market trading turnover increases at the end of the year, firms with strong green innovation capabilities have better stock performance at the end of the year. Taking Column 1 as an example, compared with October, in areas with delayed trading characteristics, the top 25% of companies in the green innovation level increased by an average of CNY 1.559 and CNY 1.908 in November and December, with a significant coefficient of 1%. Columns 2 and 4 imply no evidence that firms with weak innovation capability benefit from the delayed trade shock.

The level of green technology cannot be well measured above the median. Due to the uneven distribution of green patent indicators, we chose 75% to divide the points. Considering that Fig. 4 is analyzed according to the flow data of the company's green patent, we show the mechanism analysis under the division of the aggregation number of green patents in Table A7 and display the visualization in Fig. A7, where the results remained consistent. Given that the model setting of Table A7 also refers to Eq. (3), the coefficient interpretation of Table A7 is consistent with the above. The statement is not repeated in the text. Furthermore, an event delay may occur from the authorization of the patent to the transformation of achievements, and we use the lag of the level of green innovation to make a sound regression. The regression results are shown in Table A10.

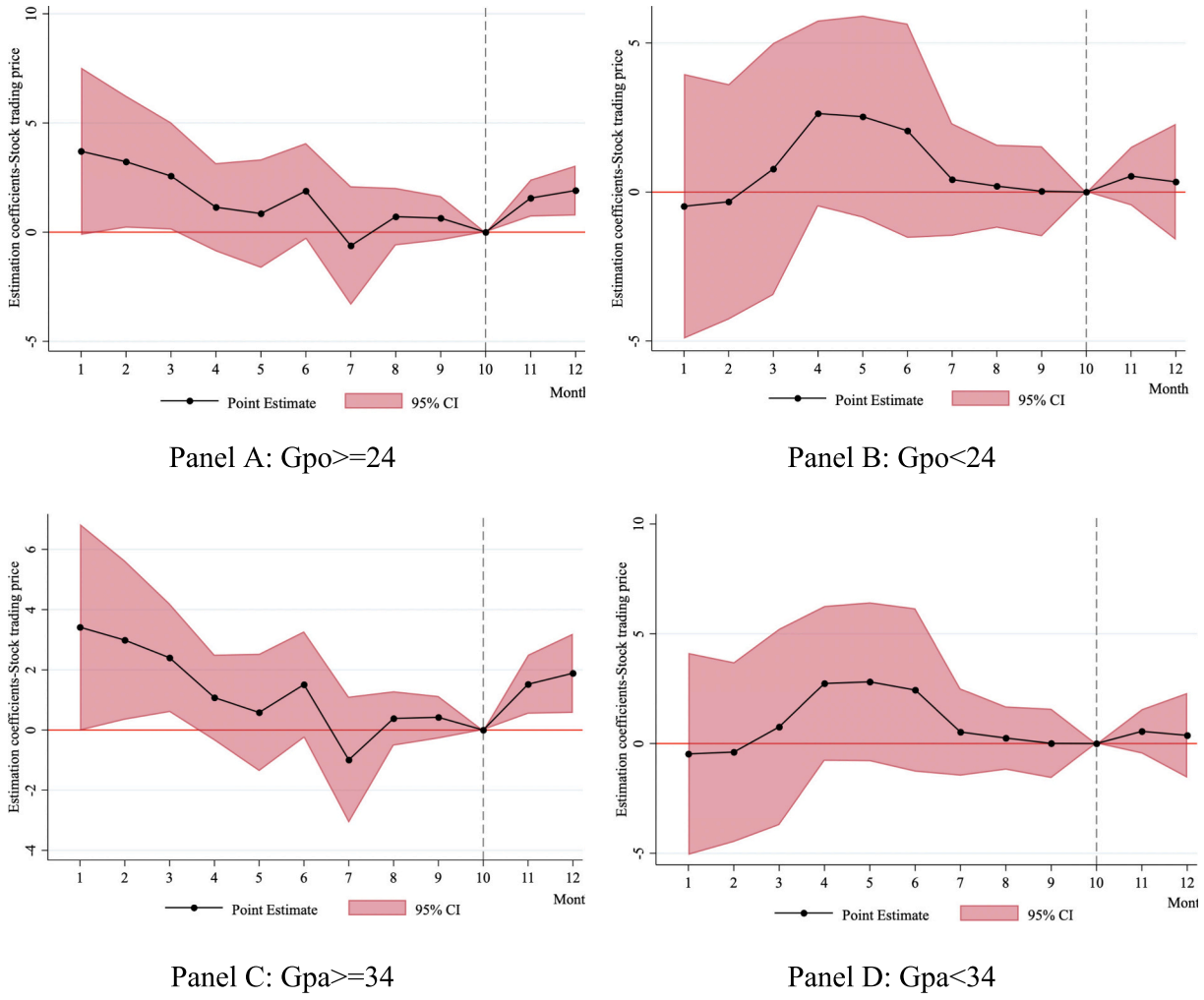



**Fig. 4.** Stock trading price trends in pilot cities.
Notes: This figure shows the time variation of trading prices in regions with the same trend of rising trading turnover in the compliance carbon market. The x-axis indicates the month. We have used the flow data of green patents to divide it. Panel A is based on firms with green patent licensing above the subsample 75% presenting value. Panel B is the variation of firms' stock prices with a low green patent acquisition. Panel C and D were divided based on green patent applications, where we classify C as firms with green firms with high patent applications, and Panel D is the regression of firms with low green patent applications. The model corresponds to Eq. (3), where we set year and firm fixed effects and explores the relationship between stock price and fulfillment transactions based on event study methods. The detailed results of the figure are shown in Table 10.

**Table 10**
Stock price performance during the fulfillment transactions period.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Gpo ≥24 | Gpo < 24 | Gpa ≥34 | Gpa < 34 |
| lead9 | 3.707* | −0.477 | 3.416* | −0.471 |
| | (1.695) | (1.880) | (1.518) | (1.946) |
| lead8 | 3.226** | −0.327 | 2.987** | −0.389 |
| | (1.335) | (1.671) | (1.169) | (1.733) |
| lead7 | 2.575** | 0.775 | 2.402** | 0.757 |
| | (1.087) | (1.788) | (0.798) | (1.892) |
| lead6 | 1.141 | 2.637* | 1.078 | 2.740 |
| | (0.894) | (1.320) | (0.631) | (1.492) |
| lead5 | 0.850 | 2.531 | 0.584 | 2.813 |
| | (1.100) | (1.434) | (0.862) | (1.531) |
| lead4 | 1.884* | 2.058 | 1.510* | 2.439 |
| | (0.973) | (1.521) | (0.783) | (1.574) |
| lead3 | −0.619 | 0.419 | −0.990 | 0.524 |
| | (1.203) | (0.799) | (0.928) | (0.841) |
| lead2 | 0.709 | 0.198 | 0.385 | 0.250 |
| | (0.582) | (0.589) | (0.400) | (0.609) |
| Lead1 | 0.642 | 0.028 | 0.424 | 0.008 |
| | (0.448) | (0.640) | (0.313) | (0.668) |
| lag1 | 1.559*** | 0.534 | 1.521*** | 0.555 |
| | (0.376) | (0.416) | (0.434) | (0.429) |
| lag2 | 1.908*** | 0.343 | 1.887** | 0.372 |
| | (0.508) | (0.826) | (0.583) | (0.819) |
| Year FEs | Y | Y | Y | Y |
| Firm FEs | Y | Y | Y | Y |
| Controls | Y | Y | Y | Y |
| Observations | 83,150 | 130,398 | 84,585 | 128,961 |
| R-squared | 0.843 | 0.650 | 0.840 | 0.653 |

Notes: This table is based on an event study approach and explores the price changes in the fulfillment transactions period for the sample that in the performance carbon trading market. We divided the companies according to the green patent authorization and application volume. Column 1 is a sample of firms with >23 green patent acquisitions, Column 2 is a sample of firms with <23 green patent acquisitions, Column 3 is a group with >31 green patent applications, and Column 4 is a group with <31 green patent applications. The results correspond to Fig. 4. The model corresponds to Eq. (3), where we set year and firm fixed effects. The standard errors are clustered at the industry-by-month level and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

### 5.2. Cost-benefit analysis

Table 11 shows the impact of carbon market trading turnover changes on firms' operating income. The results in Columns 1 and 2 reveal that an increase in carbon market transactions leads to an increase in firms' operating income with green innovations, which in turn is positively associated with an increase in stock prices. Columns 3 and 4 for non-green innovative firms imply that changes in carbon market trading do not significantly affect the firm's operating income. However, the increase in operating income still significantly affects the stock trading price. In summary, we find that companies with green innovation capabilities are better able to gain a comparative advantage in the carbon market. We classify the sample according to the number of green patent applications.

**Table 11**
Revenue mechanism analysis.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | LnTR | Clsprc | LnTR | Clsprc |
| CarbonTA | 0.013*** | | −0.003 | |
| | (0.004) | | (0.004) | |
| LnTR | | 1.262*** | | 1.182*** |
| | | (0.203) | | (0.125) |
| Year×Month FEs | Y | Y | Y | Y |
| Province FEs | Y | Y | Y | Y |
| Industry FEs | Y | Y | Y | Y |
| Controls | Y | Y | Y | Y |
| Observations | 237,095 | 240,594 | 168,083 | 168,527 |
| R-squared | 0.495 | 0.177 | 0.501 | 0.199 |

Notes: This table shows the results of the mechanism analysis of operating income. Columns 1 and 2 show the effect of the increase in carbon trading turnover on the company's revenue and the effect of the change in the revenue on the stock price for the companies with green patent acquisitions >24 (Quartile), respectively. Columns 3 and 4 show the impact of carbon trading turnover on the company's revenue and the impact of the company's operating income on the company's stock price for companies with weak green innovation. FEs means fixed effects, and the following table is the same. All models' standard errors are clustered at the industry-by-month level and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Companies with a high level of green innovation are more capable of completing emission control programs. They can use the additional carbon credits to earn profitable income and nearly increase their operating income

Although firms will increase their costs through carbon trading, for firms that are green in innovation, their respective research and development (R&D) inputs are also costs and we cannot distinguish the various sources of costs for firms. We therefore try to estimate an overall effect. The results are shown in Table 12. The results in Column 1 indicate that an increase in carbon trading has a significant positive relationship with the cost of listed companies characterized by low levels of green innovation. Column 3 implies a negative effect of cost on stock prices.

We combine the available data on the carbon emissions of listed companies to conduct a more in-depth examination of the mechanism. The carbon emission data were obtained from the Institute of Public and Environmental Affairs, a public-interest environmental research institute registered in Beijing. Owing to limited data, we combine the carbon emission data of A-share listed companies from 2018 to 2021. When matching corporate carbon emissions data, we conduct an in-depth study of the mechanism based on the full sample.

Table 13 shows the results of the sub-sample analysis. Columns 1 and 2 show the effects of the number of green patent applications and acquisitions on the carbon emissions of listed companies, respectively. The stronger the level of green innovation, the lower the amount of carbon emissions of the company, and the effect of the green patent acquisition on carbon emissions is significantly higher than the effect of the green patent application on carbon emissions. The results in Column 3 show that a decrease in carbon emissions reduces the trading turnover in the carbon market, indicating that a company's reduction in pollutant emissions reduces the demand for carbon market trading. We divide the sample into a high and low carbon emission group according to the sample median of carbon emissions.

Comparing Columns 4 and 5, we find that carbon emissions of firms with high carbon emissions have little impact on operating income, and the positive impact may come from the expansion of production scale. The pollutant emissions of companies with low carbon emissions will reduce operating income. The above results show that green innovation is an effective measure for companies to control emissions. The reduction of corporate carbon emissions will reduce the demand for carbon market trading and increase the company's operating income.

## 6. Conclusion

The change in carbon market trading turnover significantly impacts the stock market performance of carbon control companies. In the preferred specification, when carbon market trading rises by 1%, the stock prices of power generation industries and companies will fall by 0.123, equivalent to 0.656% of the average stock price level of the related industries. In addition, we focus on non-carbon control industries. None of the results is significant after replacing the sample, indicating that power generation and its related industries are more sensitive to the impact of carbon market trading. Our results also illustrate that the more relevant the industry is to the emission control policy, the more vulnerable the stock market performance is. Among the heterogeneities, the shorter the time to market and the lower the capital intensive, the more vulnerable the firms are to carbon market shocks. Firms with high green innovation capacity are affected in a statistically insignificant way.

Considering the environmental regulations of China's annual performance, we used fulfillment-based carbon trade as shocks in the context of delayed trading. The event study method shows that in four regions, carbon transactions in November and December exhibit distinct fulfillment-based trading characteristics, and often experience a notable delay towards the end of the year when compared to October. We also discussed the expectations and seasonal hybrid characteristics of the method to ensure the rationality of the design. The final results show that the start of compliance periods leads to higher trading turnover and increases the share prices of green

**Table 12**
Cost mechanism analysis.

| | (1) | (2) | (3) |
|---|---|---|---|
| | LnCost | LnCost | Clsprc |
| CarbonTA | 0.008** | 0.012 | |
| | (0.004) | (0.007) | |
| LnCost | | | −0.719*** |
| | | | (0.121) |
| Year×Month FE | Y | Y | Y |
| Province FE | Y | Y | Y |
| Industry FE | Y | Y | Y |
| Controls | Y | Y | Y |
| Observations | 136,745 | 204,467 | 344,392 |
| R-squared | 0.339 | 0.239 | 0.148 |

Notes: This table shows the results of the cost transmission mechanism. Columns 1 and 2 shows the effect of changes in trading turnover in the carbon market on firms' costs, and Column 3 shows the effect of changes in firms' costs on firms' stock prices. We logarithmize the costs. We divide the sample according to the quartile of the green patent authorization. Column 1 is the low green innovation level, and Column 2 is the high green innovation level. The model is set up to be consistent with the results of the benchmark regression. FEs means fixed effects, and the following table is the same. Standard errors are clustered at the industry-by-month level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

**Table 13**
Subsample mechanism analysis.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| | Carbon emission | Carbon emission | Carbon emission | Carbon emission | CarbonTA | LnRT | LnRT |
| Gpo | −0.225*** | | | | | | |
| | (0.042) | | | | | | |
| Gpa | | −0.187*** | | | | | |
| | | (0.046) | | | | | |
| Agg_Gpo | | | −0.010*** | | | | |
| | | | (0.003) | | | | |
| Agg_Gpa | | | | −0.007*** | | | |
| | | | | (0.002) | | | |
| Carbon emission | | | | | 0.002*** | 0.001*** | −0.041*** |
| | | | Y | Y | (0.000) | (0.000) | (0.009) |
| Year×month FEs | Y | Y | Y | Y | Y | Y | Y |
| Province FEs | Y | Y | Y | Y | Y | Y | Y |
| Industry FEs | Y | Y | Y | Y | Y | Y | Y |
| Controls | Y | Y | Y | Y | Y | Y | Y |
| Observations | 7297 | 7297 | 7297 | 7297 | 10,343 | 6300 | 6340 |
| R-squared | 0.930 | 0.931 | 0.938 | 0.938 | 0.366 | 0.931 | 0.777 |

Notes: This table shows the results of mechanism analysis for small sample data. Column 1 shows the effect of green patent acquisition on carbon emissions. Column 2 shows the effect of changes in green patent applications on carbon emissions. Column 3 reflects the effect of corporate carbon emissions on trading credits in the carbon emission market. Column 4 shows the regression results for a subsample of high-carbon emitting firms. Column 5 contains a subsample of low-carbon emitters. The threshold value is 10.96 million tons per year. FEs means fixed effects, and the following table is the same. Standard errors are clustered at the industry-by-month level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

innovative emission control companies, whereas low-green-innovative firms do not. Therefore, companies with a high level of green innovation can better adapt to the challenges brought by carbon market transactions and improve their sustainable competitiveness by adopting practical environmental protection measures and flexible business strategies. Our conclusion also implies that not all pilot areas have market characteristics of performance-based trading, which can ease the pressure of the surge in trading turnover in the carbon market at the close of the year. Companies with a weak level of green innovation should actively fulfill their contracts, prevent the expected trend of year-end transactions, and reduce the impact of the carbon market.

In the cost–benefit analysis, when the supply of carbon credits exceeds the demand, pressure on high carbon-emitting firms results in higher costs and lower profits, causing investors to expect lower share prices. Firms with high green innovation will increase the firm share price by increasing revenue. Finally, using sub-sample data, we analyzed carbon emission data of A-share listed companies and found that green innovation significantly reduces carbon emissions, while reducing pollutant emissions reduces the demand for carbon market trading and increases the operating income of companies. From the perspective of investors, decarbonization innovation can enable enterprises to maintain good stock price performance in the stock market. This discovery proves the potential benefits of taking positive actions in the field of green development. It reminds us that investors are paying more and more attention to the company's environmental protection and sustainability practices. This also provides strategic significance for enterprise decision-makers to choose between carbon market purchases and decarbonization innovation. Especially for carbon-intensive enterprises, they may need to better deal with carbon emissions and invest in innovation to maintain stock price competitiveness.

Studying the carbon market and the stock market performance of emission-controlled companies helps companies formulate carbon control policies and analyze the fluctuations of the carbon market's impact on other markets. Our conclusions have some policy orientation. First, in the carbon market, a greater diversity of firms should be included in the emission reduction scope to prevent a reduction in carbon market participation. A wider range of firms and industries can effectively alleviate the adverse investor expectations in the stock market caused by the oversupply in the carbon market. Second, companies in high carbon-emitting industries should develop timely production strategies for green innovation development. Company managers should recognize that companies with low green innovation can reduce their stock market performance owing to the undersupply of the carbon market, which may be because of the increased operating costs. Third, investors should recognize the linkage changes of the carbon market to the stock market, fully understand the complexity of the carbon market, invest cautiously, and make good use of carbon financial derivatives to diversify risks.

## CRediT authorship contribution statement

**Fanglin Chen:** Formal analysis, Methodology, Software, Validation, Writing – review & editing. **Zhongfei Chen:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing. **Xin Zhang:** Formal analysis, Methodology, Visualization, Writing – original draft.

## Data availability

The authors do not have permission to share data.

## Appendix A. Appendix

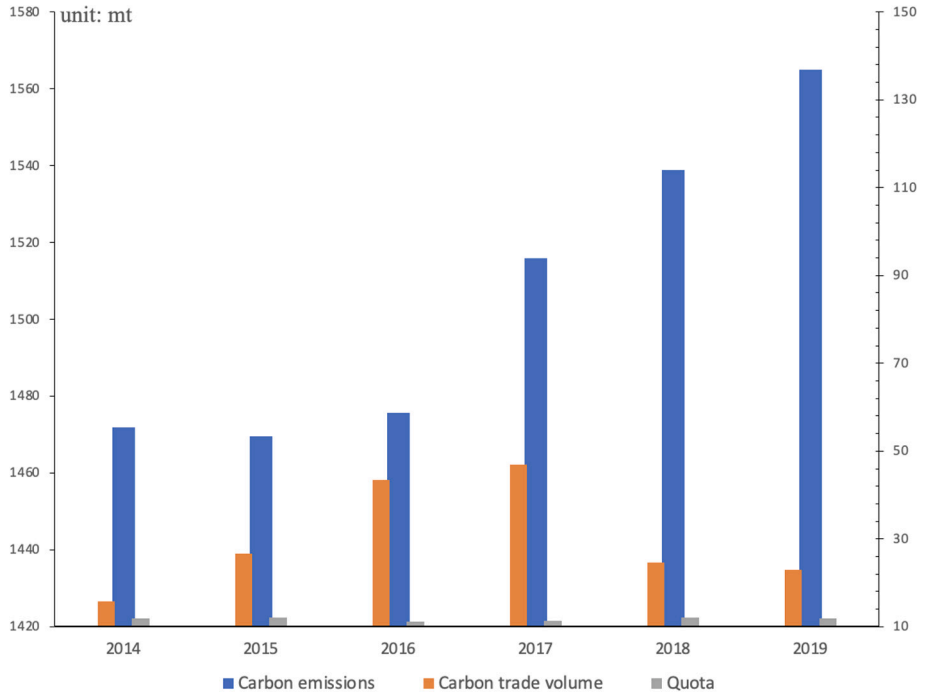



**Fig. A1.** Histogram of carbon emissions and carbon trading turnover in China's carbon pilot areas.
Notes: This figure shows the histogram of carbon emissions, carbon quotas and annual carbon transactions in China's carbon trading pilot cities from 2014 to 2019. Considering that Fujian began to pilot in 2017, we do not consider the carbon trading in the region. The data comes from the China Carbon Accounting Database, the China Carbon Emissions Trading Network, and the pilot carbon market exchanges. Regarding the treatment of carbon emissions in various regions, we refer to the treatment method of Shan et al. (2020). The yellow and gray in the figure represent the total trading turnover and total quota of each carbon market, which are placed on the sub-coordinate axis. The units of the three variables are all million tons. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

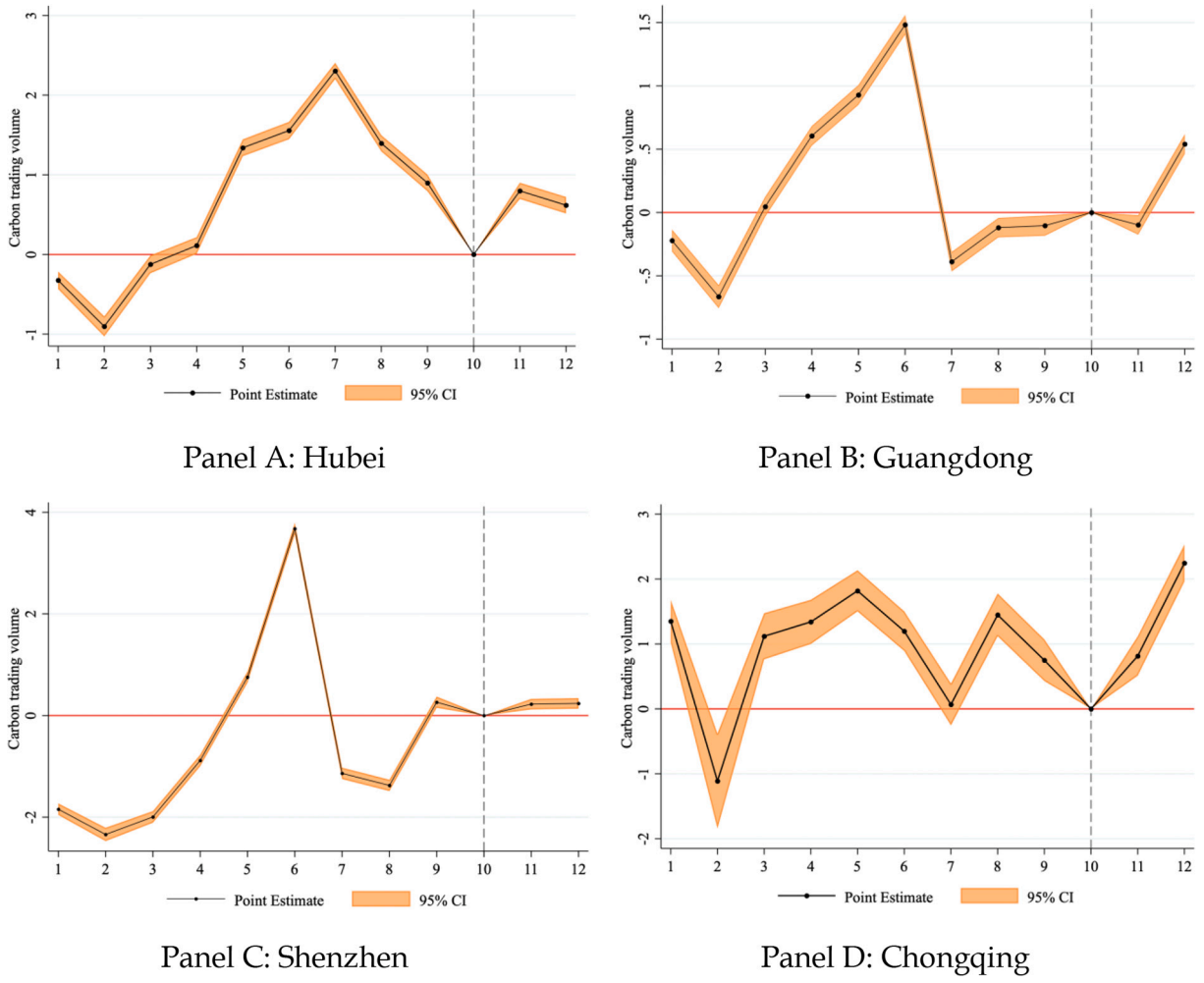



**Fig. A2.** Dynamic trend robust test for carbon trading turnovers in pilot cities.
Notes: This figure is the robustness test of Fig. 3. If too much control is added, it will lead to the omission of highly colinear variables and the loss of valid data. Therefore, we have reduced the control variables and only retained the monthly data of the purchase price index of industrial producers. The model refers to Eq. (2), with October as the shock of the performance, and the regression results are shown in Table A5.

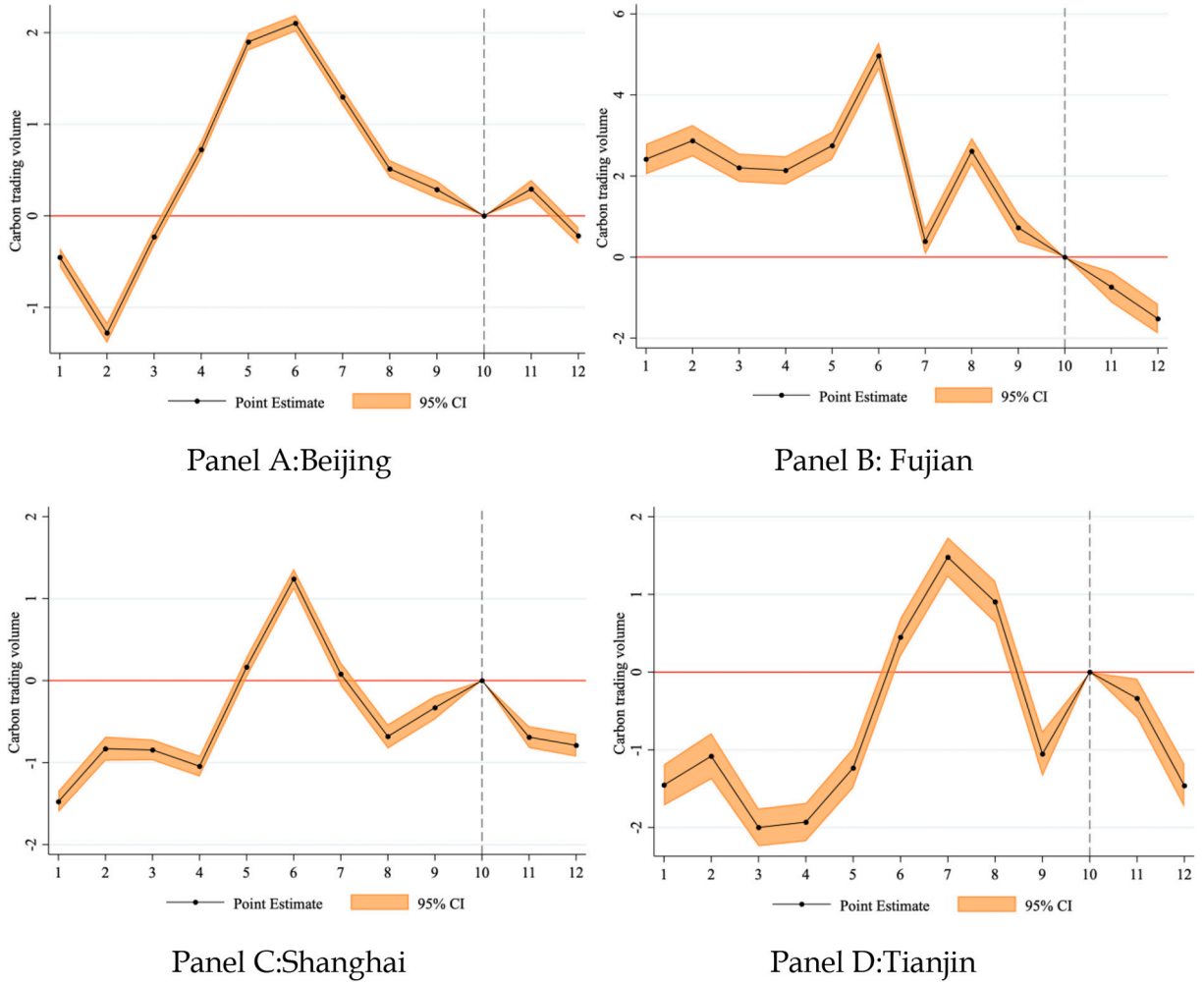



**Fig. A3.** Dynamic trend robust test for carbon trading turnovers in pilot cities (unaffected).
Notes: The figure is a supplement to Fig. 2, and the processing process is consistent. In the control variable, only the monthly data of the purchase price index of industrial producers is retained to prevent too few observations. The model setting is consistent with Eq. (2). Since we have regressed to each pilot city separately, the urban fixed effect has been removed. Taking October as the event of the approaching fulfillment transactions the regression results are displayed in Table A5.

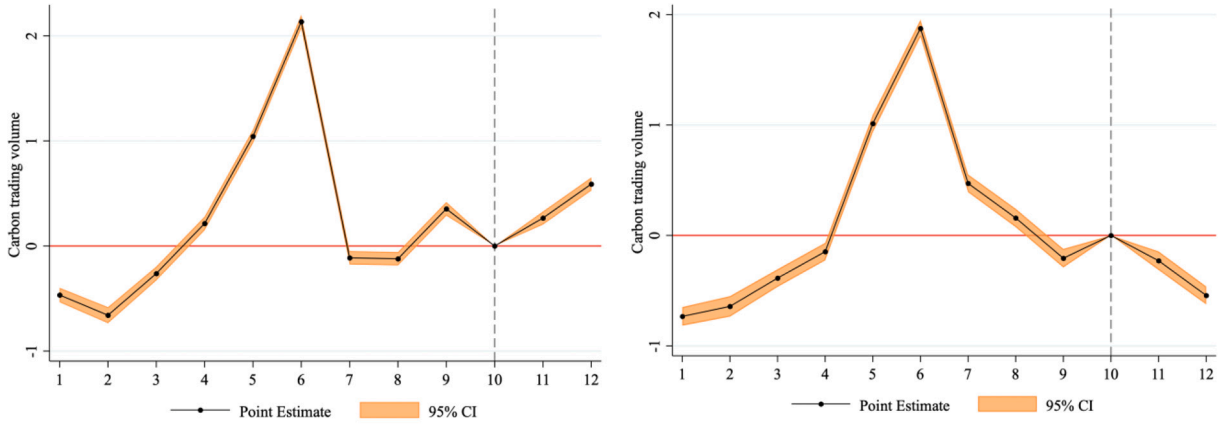



**Fig. A4.** Dynamic trend test for carbon trading turnovers in pilot cities.
Notes: The chart is the regression result of the event study model, which divides whether the carbon market will increase the carbon turnover at the end of the year, adding the fixed effect of the region and the year. Panel A contains all samples of Hubei, Guangdong, and Shenzhen Chongqing in Fig. A2. Panel B contains a sample of four performance transaction types shown in Fig. A3. The regression results are displayed in Table A6.

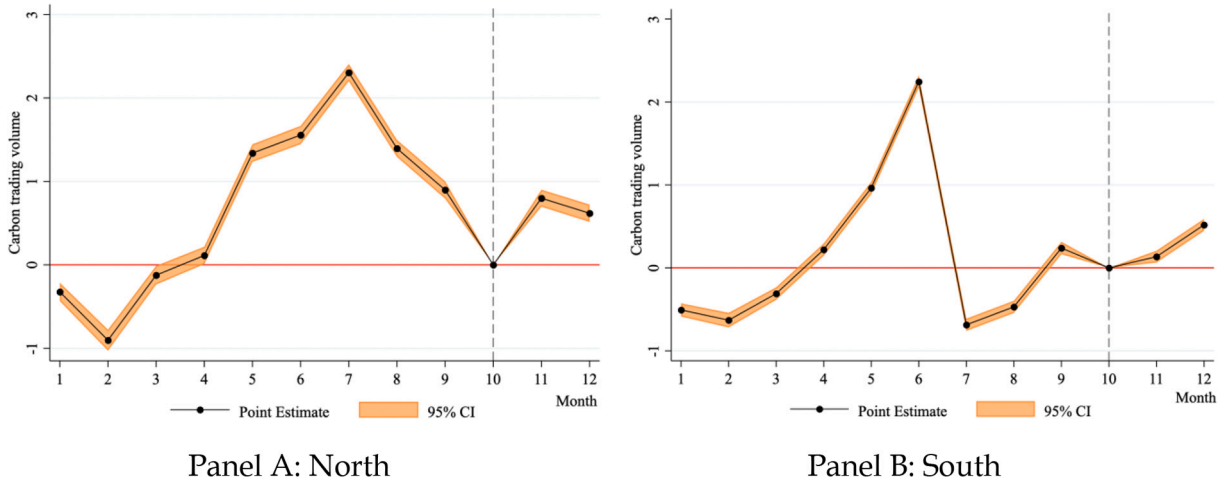



**Fig. A5.** Fulfillment-based carbon trade turnover in north-south.
Notes: This figure shows whether there are north-south differences in the total carbon transactions in areas with fulfillment-based characteristics. We divide it according to the geographical location of the pilot area. Panel A shows the dynamic trend of carbon trading turnover in the north, and Panel B shows the carbon trading trend of southern samples. The regression results of event study are shown in Table A11.

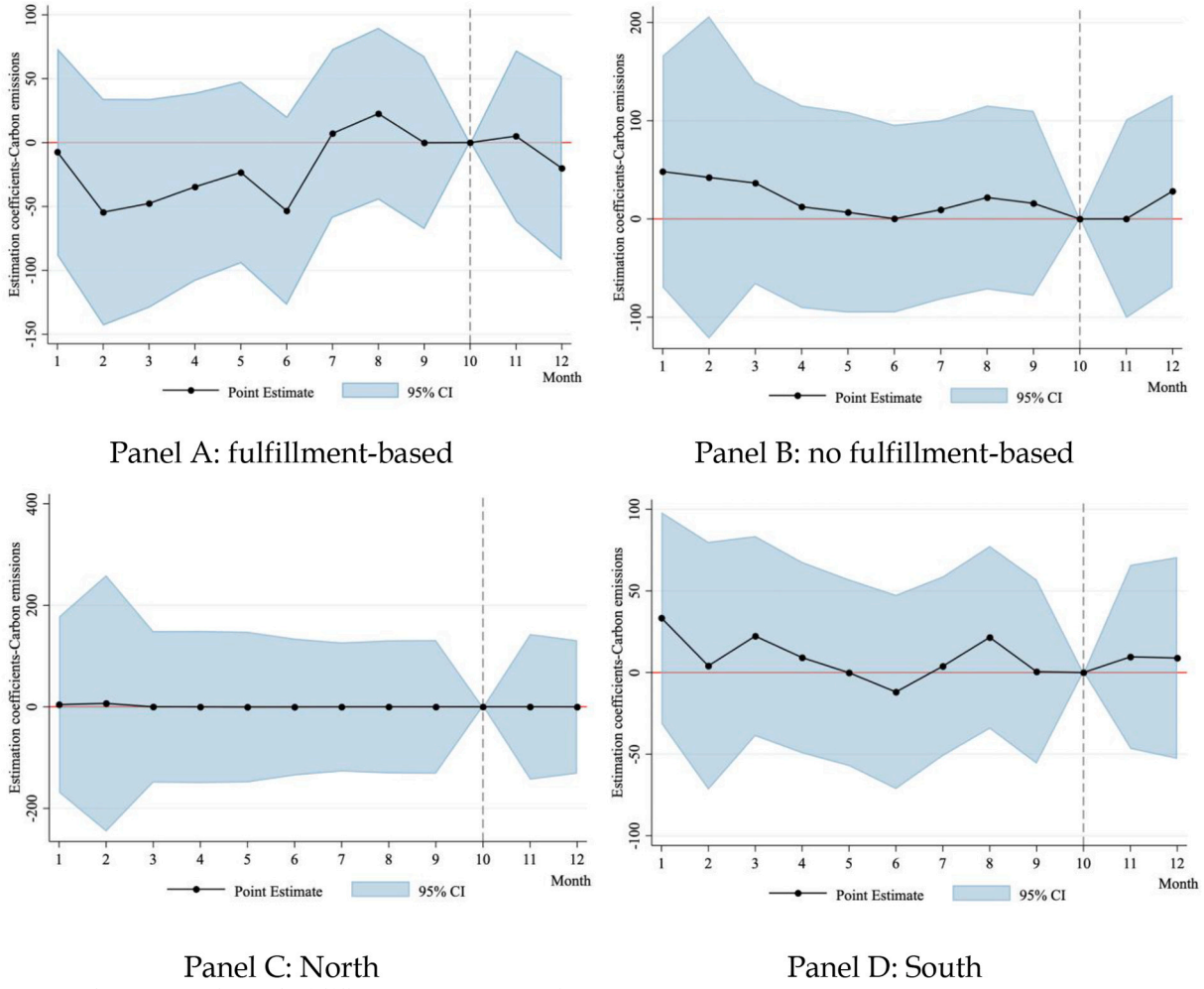



**Fig. A6.** Carbon emission during the fulfillment transactions period.

Notes: This figure is based on a small amount of data on the carbon emissions of the enterprise and explores the changes in carbon emissions by month compared to October. Panel A is for performance-oriented transactions, Panel B is for non-performance transactions, Panel C is for northern regions, and Panel D is for southern regions. Compared with October, corporate carbon emissions have no obvious seasonal characteristics. The event research design refers to Eq. (2) and replaces the explanatory variables with enterprise carbon emissions. The regression coefficient is shown in Table A12.

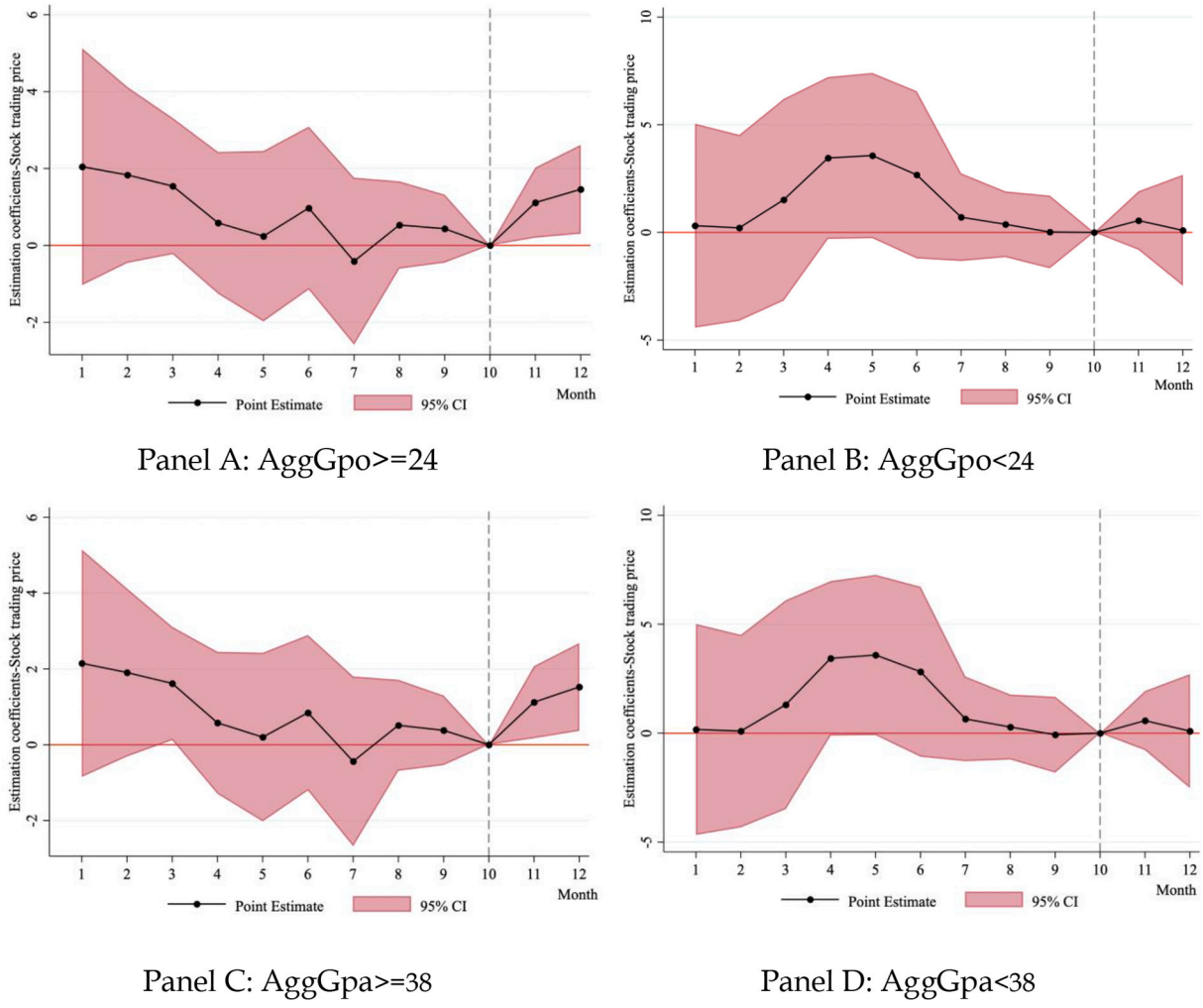



**Fig. A7.** Stock trading price trends in pilot cities (affected).
Notes: This figure shows the stock differentiation performance of companies with different green innovation capabilities during the fulfillment transactions period for performance-oriented trading regions. We use Eq. (3) to add the firm fixed effect and year fixed effect and made a sample regression according to the sample median of green patent authorization and green patent application volume. Among them, AggGpo is the authorized number of green patents, which is the stock of companies based on 2013. AggGpa is the company's green patent application stock. The regression result is consistent with Table A7. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

**Table A1**
Expanded regression of baseline.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Clsprc | Clsprc | Clsprc | Clsprc |
| CarbonTA | 0.092* | −0.028** | 0.012 | −0.014* |
| | (0.053) | (0.014) | (0.113) | (0.008) |
| Province FEs | Y | Y | Y | Y |
| Industry FEs | Y | Y | Y | Y |
| Year×month FEs | N | N | Y | Y |
| Firm FEs | Y | Y | N | N |
| Date FEs | Y | Y | N | N |
| Controls | Y | Y | Y | Y |
| Observations | 4895 | 5090 | 5024 | 5318 |
| R-squared | 0.961 | 0.983 | 0.764 | 0.980 |

Notes: This figure shows the stock differentiation performance of companies with different green innovation capabilities during the fulfillment transactions period for performance-oriented trading regions. We use Eq. (3) to add the company fixed effect and year fixed effect and made a sample regression according to the sample median of green patent authorization and green patent application volume. Among them, AggGpo is the authorized number of green patents, which is the stock of

companies based on 2013. AggGpa is the company's green patent application stock. The regression result is consistent with Table A7. FEs means fixed effects and the following table is the same. Standard errors are clustered at the industry-by-month level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

**Table A2**
Robustness test under the replacement factor model.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Clsprc | Clsprc | Clsprc | Clsprc |
| CarbonTA | −0.123*** | −0.124*** | −0.092*** | −0.092*** |
| | (0.042) | (0.042) | (0.035) | (0.035) |
| Year×month FEs | Y | Y | Y | Y |
| Province FEs | Y | Y | Y | Y |
| Industry FEs | Y | Y | Y | Y |
| Controls | Y | Y | Y | Y |
| Observations | 341,215 | 341,215 | 335,825 | 335,825 |
| R-squared | 0.123 | 0.124 | 0.190 | 0.190 |

Notes: This table shows the regression results of the replacement factor variable. Column 1 presents the benchmark regression result, which utilizes a three-factor model incorporating the market risk premium factor, market value factor, and book-to-market value ratio factor. These factors are weighted based on the circulating market value. The three-factor index is replaced by the total market value-weighted calculation in Column 2. Column 3 is the five-factor model that additionally considers the profitability factor and the investment model factor, weighted according to the circulating market value. Column 4 replaces the five-factor data with a variable weighted by the total market value. FEs means fixed effects and the following table is the same. Standard errors are clustered at the industry-by-month level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

**Table A3**
Robustness tests.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| | Clsprc | Clsprc | Clsprc | Clsprc | Clsprc | Clsprc | Clsprc | Clsprc |
| CarbonTA | −0.123** | −0.123** | −0.123** | −0.136** | −0.125** | −0.108*** | −0.121*** | −0.123*** |
| | (0.041) | (0.041) | (0.041) | (0.049) | (0.039) | (0.034) | (0.043) | (0.041) |
| Year×Month FEs | Y | Y | Y | Y | Y | Y | Y | Y |
| Province FEs | Y | Y | Y | Y | Y | | | Y |
| Industry FEs | Y | Y | Y | Y | Y | Y | Y | Y |
| Week FEs | | Y | | | | | | |
| Dow FEs | | | Y | | | | | |
| Days FEs | | | | Y | | | | |
| Province×Year FEs | | | | | | Y | | |
| City×Year FEs | | | | | | | Y | |
| Controls | Y | Y | Y | Y | Y | Y | Y | Y |
| Observations | 341,215 | 341,215 | 341,215 | 341,215 | 325,112 | 341,215 | 341,209 | 341,215 |
| R-squared | 0.123 | 0.123 | 0.123 | 0.120 | 0.122 | 0.129 | 0.137 | 0.123 |

Notes: This table presents the regression results for the robustness tests. Column 1 is the baseline regression, and Columns 2–4 add Week, Dow, and Days fixed effects respectively. Column 5 excludes the stock market crash period sample (June 2015–January 2016). Column 6 uses the province by year fixed effects. City by year fixed effect is used in Column 7. Column 8 changes the cluster stand error. We use a more relaxed condition, which adopted the industry-year-month clustering standard error. FEs means fixed effects and the following table is the same. Standard errors are clustered at the industry-by-month level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

**Table A4**
Heterogeneity analysis: private and state-owned enterprises.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Clsprc | Clsprc | Clsprc | Clsprc |
| CarbonTA | −0.123*** | 0.073*** | −0.267*** | −0.025 |
| | (0.042) | (0.027) | (0.050) | (0.046) |
| CarbonTA#Private | | | | −0.166*** |
| | | | | (0.051) |
| Year×Month FEs | Y | Y | Y | Y |
| Province FEs | Y | Y | Y | Y |
| Industry FEs | Y | Y | Y | Y |
| Controls | Y | Y | Y | Y |
| Observations | 341,215 | 125,602 | 215,612 | 341,215 |
| R-squared | 0.123 | 0.429 | 0.112 | 0.125 |

Notes: This table presents the results of a regression of the heterogeneity effect of carbon trading on stock prices. We divide them according to whether they are privately owned or not. Column 1 presents the results of the baseline regression. Column 2 shows the

results of the regression for state-owned enterprises, and Column 3 shows the regression for a sub-sample of private enterprises. In Column 4, we added the dummy variable of whether it is a private enterprise and its interaction with the variable CarbonTA. FEs means fixed effects and the following table is the same. Standard errors are clustered at the industry-by-month level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

**Table A5**
Robust test of the fulfillment transactions event shock.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| | Beijing | Fujian | Shanghai | Tianjin | Hubei | Guangdong | Shenzhen | Chongqing |
| Lead9 | −0.452*** | 2.416*** | −1.476*** | −1.454*** | −0.324*** | −0.221*** | −1.842*** | 1.350*** |
| | (0.053) | (0.191) | (0.067) | (0.136) | (0.056) | (0.044) | (0.058) | (0.167) |
| Lead8 | −1.278*** | 2.870*** | −0.831*** | −1.083*** | −0.903*** | −0.665*** | −2.340*** | −1.111*** |
| | (0.058) | (0.198) | (0.075) | (0.152) | (0.064) | (0.046) | (0.067) | (0.373) |
| Lead7 | −0.229*** | 2.205*** | −0.845*** | −1.999*** | −0.124** | 0.046 | −1.994*** | 1.118*** |
| | (0.050) | (0.179) | (0.065) | (0.124) | (0.057) | (0.040) | (0.058) | (0.181) |
| Lead6 | 0.724*** | 2.139*** | −1.044*** | −1.931*** | 0.113** | 0.606*** | −0.884*** | 1.340*** |
| | (0.049) | (0.178) | (0.065) | (0.126) | (0.053) | (0.040) | (0.058) | (0.174) |
| Lead5 | 1.899*** | 2.749*** | 0.165** | −1.235*** | 1.339*** | 0.929*** | 0.756*** | 1.818*** |
| | (0.048) | (0.177) | (0.065) | (0.132) | (0.054) | (0.040) | (0.056) | (0.162) |
| Lead4 | 2.103*** | 4.966*** | 1.240*** | 0.449*** | 1.556*** | 1.483*** | 3.678*** | 1.197*** |
| | (0.047) | (0.173) | (0.065) | (0.127) | (0.056) | (0.039) | (0.058) | (0.156) |
| Lead3 | 1.298*** | 0.389** | 0.081 | 1.479*** | 2.304*** | −0.389*** | −1.137*** | 0.068 |
| | (0.047) | (0.169) | (0.073) | (0.131) | (0.052) | (0.039) | (0.057) | (0.164) |
| Lead2 | 0.513*** | 2.612*** | −0.681*** | 0.905*** | 1.395*** | −0.120*** | −1.374*** | 1.448*** |
| | (0.049) | (0.171) | (0.076) | (0.138) | (0.052) | (0.039) | (0.057) | (0.168) |
| Lead1 | 0.287*** | 0.726*** | −0.329*** | −1.053*** | 0.898*** | −0.104** | 0.263*** | 0.750*** |
| | (0.050) | (0.178) | (0.074) | (0.149) | (0.053) | (0.041) | (0.057) | (0.164) |
| Lag1 | 0.293*** | −0.737*** | −0.689*** | −0.338*** | 0.799*** | −0.098** | 0.226*** | 0.814*** |
| | (0.052) | (0.196) | (0.068) | (0.130) | (0.052) | (0.039) | (0.053) | (0.155) |
| Lag2 | −0.217*** | −1.519*** | −0.789*** | −1.462*** | 0.618*** | 0.541*** | 0.239*** | 2.246*** |
| | (0.048) | (0.188) | (0.071) | (0.144) | (0.053) | (0.040) | (0.053) | (0.146) |
| Year FEs | Y | Y | Y | Y | Y | Y | Y | Y |
| Controls | Y | Y | Y | Y | Y | Y | Y | Y |
| Observations | 62,779 | 14,473 | 55,569 | 4512 | 36,004 | 96,229 | 60,067 | 7257 |
| R-squared | 0.218 | 0.295 | 0.074 | 0.560 | 0.257 | 0.244 | 0.448 | 0.132 |

Notes: This table shows the regression results of the event trend coefficient of carbon trading turnover in each pilot area. On the basis of Eq. (2), we divide the sample regression. Compared with Table 9, we have reduced the number of control variables to prevent missing variables under multiple collinearities and increased the number of effective observations. FEs means fixed effects and the following table is the same. Standard errors are clustered at the industry-by-month level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. The regression results correspond to Fig. A2 and Fig. A3.

**Table A6**
Dynamic trend test of regions with different carbon trading characteristics.

| | (1) | (2) |
|---|---|---|
| | CarbonTA | CarbonTA |
| Lead9 | −0.467*** | −0.733*** |
| | (0.035) | (0.043) |
| Lead8 | −0.659*** | −0.643*** |
| | (0.039) | (0.047) |
| Lead7 | −0.262*** | −0.386*** |
| | (0.034) | (0.041) |
| Lead6 | 0.215*** | −0.147*** |
| | (0.033) | (0.041) |
| Lead5 | 1.043*** | 1.013*** |
| | (0.033) | (0.040) |
| Lead4 | 2.135*** | 1.875*** |
| | (0.033) | (0.039) |
| Lead3 | −0.113*** | 0.472*** |
| | (0.033) | (0.041) |
| Lead2 | −0.122*** | 0.158*** |
| | (0.033) | (0.042) |
| Lead1 | 0.354*** | −0.206*** |
| | (0.033) | (0.043) |
| Lag1 | 0.266*** | −0.229*** |
| | (0.032) | (0.043) |
| Lag2 | 0.589*** | −0.543*** |
| | (0.032) | (0.041) |

(*continued on next page*)

**Table A6** (*continued*)

| | (1) | (2) |
|---|---|---|
| | CarbonTA | CarbonTA |
| Year FEs | Y | Y |
| Area FEs | Y | Y |
| Controls | Y | Y |
| Observations | 199,557 | 137,333 |
| R-squared | 0.232 | 0.142 |

Notes: This table corresponds to the dynamic trend of carbon trading turnover in performance and non-fulfilled pilot areas. Column 1 is the regression results of four pilot areas with the nature of performance transactions (the fulfillment transactions will increase significantly in carbon market transactions). Column 2 is the dynamic trend test of the total carbon trading turnover in the four pilot areas of non-performance transactions. The model is consistent with Eq. (2). FEs means fixed effects and the following table is the same. The visualization of regression results corresponds to Fig. A4.

**Table A7**
Stock price performance during the fulfillment transactions period.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | AggGpo≥ 24 | AggGpo<24 | AggGpa≥ 38 | AggGpa<38 |
| lead9 | 2.047 | 0.312 | 2.150 | 0.170 |
| | (1.362) | (2.004) | (1.326) | (2.047) |
| lead8 | 1.832 | 0.211 | 1.906* | 0.094 |
| | (1.011) | (1.825) | (0.980) | (1.868) |
| lead7 | 1.542* | 1.514 | 1.617** | 1.308 |
| | (0.782) | (1.981) | (0.661) | (2.026) |
| lead6 | 0.585 | 3.456* | 0.579 | 3.434* |
| | (0.815) | (1.589) | (0.828) | (1.500) |
| lead5 | 0.240 | 3.568* | 0.201 | 3.587* |
| | (0.980) | (1.622) | (0.983) | (1.555) |
| lead4 | 0.972 | 2.673 | 0.844 | 2.820 |
| | (0.936) | (1.642) | (0.906) | (1.650) |
| lead3 | −0.412 | 0.708 | −0.437 | 0.656 |
| | (0.960) | (0.861) | (0.991) | (0.823) |
| lead2 | 0.531 | 0.376 | 0.512 | 0.282 |
| | (0.502) | (0.645) | (0.531) | (0.630) |
| Lead1 | 0.436 | 0.020 | 0.381 | −0.071 |
| | (0.390) | (0.714) | (0.406) | (0.736) |
| lag1 | 1.113** | 0.549 | 1.123** | 0.577 |
| | (0.403) | (0.574) | (0.420) | (0.574) |
| lag2 | 1.459** | 0.095 | 1.525** | 0.094 |
| | (0.511) | (1.089) | (0.512) | (1.107) |
| Year FEs | Y | Y | Y | Y |
| Firm FEs | Y | Y | Y | Y |
| Controls | Y | Y | Y | Y |
| Observations | 126,435 | 87,113 | 125,700 | 87,844 |
| R-squared | 0.801 | 0.704 | 0.818 | 0.699 |

Notes: This table shows the monthly fluctuation of stock trading prices, and we set October as a reference group. It is based on an event study approach and explores the price changes in the fulfillment transactions period for the sample with the same trading trend in the fulfillment transactions event. We divided the company's samples according to the green patent application stock and authorized stock. Column 1 is a sample of firms with more than the median of green patent obtain (Gpo), Column 2 is a sample of firms with less than the median of aggerate Gpo, Column 3 is a group with more than the median of aggerate green patent applications (Gpa), and Column 4 is a group with less than the median of aggerate green patent applications. The results correspond to Fig. 4. All models include the control of the company's age. Standard errors are clustered at the year level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

**Table A8**
Heterogeneity analysis of innovative applications.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | L_Gpo | H_Gpo | Interaction | L_Gpa | H_Gpa | Interaction |
| CarbonTA | −0.091*** | 0.064 | −0.242*** | −0.104*** | 0.075** | −0.249*** |
| | (0.027) | (0.039) | (0.049) | (0.028) | (0.037) | (0.050) |
| CarbonTA×H_Gpo | | | 0.182** | | | |

(*continued on next page*)

**Table A8** (*continued*)

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | L_Gpo | H_Gpo | Interaction | L_Gpa | H_Gpa | Interaction |
| | | | (0.071) | | | |
| CarbonTA×H_Gpa | | | | | | 0.217*** |
| | | | | | | (0.075) |
| Year×Month FE | Y | Y | Y | Y | Y | Y |
| Province FE | Y | Y | Y | Y | Y | Y |
| Industry FE | Y | Y | Y | Y | Y | Y |
| Controls | Y | Y | Y | Y | Y | Y |
| Observations | 205,069 | 68,651 | 341,215 | 204,418 | 69,302 | 341,215 |
| R-squared | 0.209 | 0.426 | 0.154 | 0.209 | 0.492 | 0.154 |

Notes: This table shows the regression results of the impact of carbon trading on the stock prices of companies with different levels of green innovation. Compared with Table 8, this table uses the stock of green patent applications as the basis for division. The structural index H_Gpa indicates whether the company's green innovation application level is in the top 25% of the industry. Among them, Column 1 is the sample of enterprises with weak green innovation ability, Column 2 corresponds to the enterprises with strong green innovation ability, Column 3 added the H_Gpa and interaction between dummy and core explanatory variables. FEs means fixed effects and the following table is the same. Standard errors are clustered at the industry-by-month level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

**Table A9**
Heterogeneity analysis: time of listing.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | S_Time | L_Time | Interaction | S_Time | L_Time | Interaction |
| CarbonTA | −0.533*** | 0.088*** | −0.348*** | −0.227*** | 0.020 | −0.198*** |
| | (0.124) | (0.029) | (0.125) | (0.044) | (0.022) | (0.045) |
| CarbonTA#L_T1 | | | 0.294** | | | |
| | | | (0.143) | | | |
| CarbonTA#L_T2 | | | | | | 0.174*** |
| | | | | | | (0.051) |
| Year×Month FE | Y | Y | | Y | Y | Y |
| Province FE | Y | Y | | Y | Y | Y |
| Industry FE | Y | Y | | Y | Y | Y |
| Controls | Y | Y | | Y | Y | Y |
| Observations | 78,560 | 262,655 | 341,215 | 252,750 | 88,463 | 341,215 |
| R-squared | 0.252 | 0.285 | 0.181 | 0.138 | 0.314 | 0.128 |

Notes: This table shows the regression results of the impact of carbon trading on the stock prices of companies with different levels of firm's listing time. Compared with Table 7, This table uses the upper and lower quartile (L_T1:4-year, L_T2: 18-year) of listing time as the standard for dividing subsamples and performs the regression of interactive items. FEs means fixed effects and the following table is the same. Standard errors are clustered at the industry-by-month level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

**Table A10**
Stock price performance during the fulfillment transactions period.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Lag_Gpo | Lag_Gpo | Lag_Gpa | Lag_Gpa |
| lead9 | 4.213 | 0.341 | 4.331* | 0.247 |
| | (2.316) | (1.772) | (2.343) | (1.723) |
| lead8 | 3.803* | 0.669 | 4.014** | 0.525 |
| | (1.797) | (1.381) | (1.729) | (1.415) |
| lead7 | 3.105* | 1.381 | 3.299** | 1.251 |
| | (1.406) | (1.455) | (1.382) | (1.506) |
| lead6 | 0.877 | 2.321 | 1.073 | 2.188 |
| | (0.927) | (1.476) | (1.129) | (1.504) |
| lead5 | 0.814 | 2.242 | 1.173 | 2.078 |
| | (1.303) | (1.534) | (1.616) | (1.501) |
| lead4 | 1.806 | 2.075 | 2.076 | 1.915 |
| | (1.179) | (1.414) | (1.433) | (1.348) |
| lead3 | −0.564 | 0.224 | 0.004 | 0.054 |
| | (1.246) | (1.063) | (1.286) | (1.096) |
| lead2 | −0.036 | 0.584 | 0.317 | 0.498 |
| | (1.078) | (0.645) | (1.085) | (0.657) |
| Lead1 | 0.235 | 0.395 | 0.437 | 0.322 |
| | (0.864) | (0.565) | (0.845) | (0.583) |
| lag1 | 1.780*** | 0.637 | 1.617** | 0.664 |
| | (0.485) | (0.449) | (0.560) | (0.434) |

(*continued on next page*)

**Table A10** (*continued*)

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Lag_Gpo | Lag_Gpo | Lag_Gpa | Lag_Gpa |
| lag2 | 1.698* | 0.645 | 1.646* | 0.657 |
| | (0.800) | (0.702) | (0.812) | (0.684) |
| Year FEs | Y | Y | Y | Y |
| Firm FEs | Y | Y | Y | Y |
| Controls | Y | Y | Y | Y |
| Observations | 66,764 | 147,793 | 67,353 | 147,204 |
| R-squared | 0.706 | 0.675 | 0.709 | 0.671 |

Notes: This table is a robust supplement to Table A7, which the model settings are consistent. The difference is that considering the lag of green innovation. FEs means fixed effects and the following table is the same. Standard errors are clustered at the year level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

**Table A11**
Differences in North-South carbon trading turnovers.

| | (1) | (2) |
|---|---|---|
| | North | South |
| Lead9 | −0.324*** | −0.507*** |
| | (0.056) | (0.041) |
| Lead8 | −0.903*** | −0.630*** |
| | (0.064) | (0.045) |
| Lead7 | −0.124** | −0.310*** |
| | (0.057) | (0.038) |
| Lead6 | 0.113** | 0.220*** |
| | (0.053) | (0.038) |
| Lead5 | 1.339*** | 0.963*** |
| | (0.054) | (0.038) |
| Lead4 | 1.556*** | 2.244*** |
| | (0.056) | (0.038) |
| Lead3 | 2.304*** | −0.686*** |
| | (0.052) | (0.038) |
| Lead2 | 1.395*** | −0.470*** |
| | (0.052) | (0.038) |
| Lead1 | 0.898*** | 0.239*** |
| | (0.053) | (0.038) |
| Lag1 | 0.799*** | 0.135*** |
| | (0.052) | (0.037) |
| Lag2 | 0.618*** | 0.518*** |
| | (0.053) | (0.037) |
| Year FEs | Y | Y |
| Area FEs | Y | Y |
| Controls | Y | Y |
| Observations | 36,004 | 163,553 |
| R-squared | 0.257 | 0.218 |

Notes: This table shows whether there is a north-south difference in carbon trading turnovers in areas with fulfillment-based transactions. Our results are visualized in Fig. A5. Among them, Column 1 corresponds to the north and Column 2 corresponds to the southern region. The model setting is consistent with Eq. (2). FEs means fixed effects and the following table is the same. Standard errors are clustered at the year level in all models and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

**Table A12**
Carbon emission during the fulfillment transactions period.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Carbon emissions | | | |
| lead9 | −7.341 | 48.267 | 4.533 | 33.415 |
| | (41.401) | (60.193) | (88.549) | (33.111) |
| lead8 | −54.505 | 42.271 | 6.816 | 4.091 |
| | (45.251) | (83.851) | (128.871) | (38.747) |
| lead7 | −47.631 | 36.547 | 0.070 | 22.322 |
| | (41.649) | (52.594) | (76.168) | (31.278) |
| lead6 | −34.670 | 12.391 | −0.195 | 9.146 |

(*continued on next page*)

**Table A12** (*continued*)

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Carbon emissions | | | |
| | (37.544) | (52.639) | (76.482) | (29.906) |
| lead5 | −23.372 | 6.714 | −0.448 | −0.130 |
| | (36.265) | (52.128) | (75.722) | (29.170) |
| lead4 | −53.379 | 0.259 | −0.347 | −11.916 |
| | (37.653) | (48.736) | (68.777) | (30.358) |
| lead3 | 7.169 | 9.418 | −0.200 | 3.850 |
| | (33.702) | (46.603) | (64.859) | (28.074) |
| lead2 | 22.616 | 21.859 | −0.070 | 21.541 |
| | (34.293) | (47.770) | (66.924) | (28.587) |
| Lead1 | −0.118 | 15.864 | −0.100 | 0.527 |
| | (34.551) | (48.028) | (67.190) | (28.797) |
| lag1 | 5.056 | 0.090 | 0.021 | 9.591 |
| | (34.286) | (51.581) | (73.219) | (28.752) |
| lag2 | −20.043 | 28.275 | −0.141 | 8.919 |
| | (36.758) | (50.040) | (67.061) | (31.560) |
| Year FEs | Y | Y | Y | Y |
| Area FEs | Y | Y | Y | Y |
| Controls | Y | Y | Y | Y |
| Observations | 2453 | 6405 | 4353 | 4505 |
| R-squared | 0.120 | 0.184 | 0.044 | 0.101 |

Notes: This table shows the monthly fluctuation of carbon emission, and we set October as a reference group. It is based on an event study approach and explores the north-south difference in the fulfillment transactions period. Column 1 is the trend of carbon emissions of companies in the pilot areas for fulfillment-based trading, Column 2 is for non-performance pilot areas, Column 3 is the northern region, and Column 4 is the southern region. The visualization of regression results is shown in Fig. A6.

**Table A13**
Recursive feature elimination: weather higher.

| Order | Index | Order | Index |
|---|---|---|---|
| 1 | Is Hubei | 4 | Is Guangdong |
| 2 | Is Tianjin | 5 | Is Shenzhen |
| 3 | Is Chongqing | 6 | Carbon trading turnover |

Notes: This table is the factor that most affect carbon trading turnovers obtained by using XGBoost integration algorithm and feature screening. Geographic variables are indicator variables that convert classification variables in pilot areas through one-hot code. We use random forests to fill in the missing values. If there are vacant values in multiple columns, first sort out the amount of missing data in each column and fill them from the column with the least missing amount. The fewer the missing values, the lower the information entropy, the easier it is to process, and the higher the accuracy.

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